Estimating Local Average Treatment Effects Under Weak Monotonicity

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Abstract

We study the estimation of treatment effects with an instrumental variable when there are compliers and defiers. If the two groups exist at distinct covariate values, then the instrumental variable identifies a local average treatment effect, but conventional two-stage least squares does not estimate it. An oracle with knowledge of the defier covariate values could correct the two-stage least squares estimate by applying a sign correction wherever defiers are present. We propose a feasible version of this estimator where estimated propensity scores are used to identify defier covariate values. We characterise the asymptotic distribution of the feasible estimator and show conditions under which it is asymptotically equivalent to the oracle estimator. We provide a consistent estimator for the standard error of the feasible estimator, allowing inference. We evaluate the finite sample properties of the feasible estimator by simulation, and find that it outperforms the oracle estimator in terms of mean squared error. We apply the estimator to the data of Card (1993). Compared to conventional two-stage least squares estimates, the point estimates obtained by our estimator differ in sign and are insignificant. These results suggest that the instrument of Card (1993) is weak.

1 Introduction

Instrumental variables generate exogenous variation in a variable of interest while having no direct causal effect on the outcome. In the heterogeneous treatment effect model, variation arises as the instrument affects the treatment decisions of a group of switchers. Switchers may be compliers, who are induced to opt-in to treatment by the instrument, or defiers, who are induced to opt-out.

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When both types are present, the instrument induces flows in and out of treatment. If compliers cannot be distinguished from defiers, then variation in the instrument identifies some mixture of the effect of treatment take-up for compliers and treatment drop-out for defiers, which may not correspond to the treatment effect experienced by anyone in the population. In a setting without covariates, Imbens and Angrist (1994) resolves this by adopting the monotonicity assumption: only one type of switcher exists. The instrumental variable then identifies an average of the treatment effects experienced by that group, and this average can be estimated by two-stage least squares.

In the heterogeneous treatment effect model with covariates, the instrumental variable identifies an average treatment effect if monotonicity holds conditional on covariates. Both compliers and defiers may be present, but not in the same covariate stratum. Słoczyński (2026) christens this 'weak monotonicity'. In this setting, the type of switcher is identified by the probability of treatment conditional on covariates and the instrument. That is, the propensity score. For a given covariate stratum, if switchers are compliers, then the propensity score is increasing with the instrument. If they are defiers, it is decreasing. This allows an average treatment effect to be estimated for each group and, in turn, an overall average.

Słoczyński (2026) points out that, under weak monotonicity, the workhorse two-stage least squares specification used in empirical work does not estimate a weighted average of treatment effects. Heckman and Vytlacil (2005) and Kolesár (2013) show that, for the two-stage least squares estimand to correspond to a weighted average of treatment effects, the first stage must accurately approximate the effect of the instrument on the propensity score insofar as the sign is correct: a change in the instrument which increases the propensity score must also increase the first stage prediction for treatment. The standard two-stage least squares specification omits covariate-instrument interactions. In doing so, it imposes that the sign of the effect of instrument on the propensity score is constant. With two types of switchers, this does not hold. As a result, one type of switcher receives negative weights in the two-stage least squares estimand.

In response to this Słoczyński (2026), advocates the saturate and control specification of Angrist and Imbens (1995), where covariates are converted to a single categorical variable that is interacted with the instrument. This specification essentially estimates a separate first stage regression for each covariate stratum. As such, it places no restrictions on the sign of the first stage estimate across covariate strata. However, if there are few observations in each stratum, then it can produce weak instruments. Furthermore, it does not readily accommodate continuous covariates. Słoczyński (2026) notes that the weighted average interpretation of the two-stage least squares estimand can be restored by reordering the instrument. Defiers for an instrument Z are compliers for the instrument $1 - Z$. Hence, an oracle with knowledge of switcher types among covariate strata could construct an instrument such that all switchers are compliers by reordering the instrument on a cell-by-cell

basis. Słoczyński (2026) shows that this reordered IV estimator identifies a weighted average of treatment effects, but leaves the investigation of its properties to future work.

In this paper, we consider a feasible reordered IV estimator. As switcher type is identified by propensity scores, we use the estimated propensity score function to classify switchers. Given a generic estimator of the propensity score, we show conditions under which the feasible reordered IV estimator is asymptotically equivalent to an oracle estimator with full knowledge of switcher types. This is true under two conditions. First, the propensity score must be estimated at a sufficiently fast rate. Second, a margin condition, which governs the density of covariate strata where the instrument is weak, must hold. The feasible and oracle estimators differ when there exist covariate strata where the feasible estimator misclassifies switchers. Asymptotically, classification mistakes are concentrated in strata where switchers are hardest to classify. As switcher type is identified by the effect of the instrument on the propensity score, these strata are those where the instrument has little effect. Strata where the instrument is close to irrelevant also receive small weights in two-stage least squares type estimands. Hence, asymptotic classification errors have little effect on the estimand.

To classify switchers, we require only knowledge of the sign of the effect of the instrument on the propensity score. We leverage this to simplify estimation. We specify the propensity score as a nonparametric strictly increasing transformation of a linear index. As a strictly increasing transformation does not affect rank, we need only estimate the parameters of the linear index. We show that the propensity score has this form if the random assignment assumption of Imbens and Angrist (1994) is strengthened and an additional functional form restriction is imposed. This semiparametric specification is more flexible than assuming a parametric model for the propensity score, but the parametric component can be estimated at the $\sqrt{n}$-rate under mild restrictions. We propose estimating the parameters of the linear index using the maximum rank correlation estimator of Han (1987). This approach does not directly estimate the nonparametric function, allowing us to sidestep bandwidth selection.

We show that reordered IV with a semiparametric plug-in is $\sqrt{n}$ consistent and asymptotically normal, and is oracle equivalent when the margin condition holds. In addition, we develop an easy to implement estimator of its asymptotic standard error. To characterise the finite sample properties of our estimator, we conduct a simulation exercise. We consider two processes. The first is specified such that weak monotonicity holds, and the proportions of compliers and defiers are similar. For the second, strong monotonicity holds, but compliers account for only a small proportion of the population so the instrument is weak. We compare the feasible and oracle estimators. Consistent with our theoretical results, as the sample size increases, their sampling distributions converge. However, in smaller sample sizes, the variance of the sampling distribution of the feasible estimator

is far lower than the oracle estimator. This leads the feasible estimator to outperform the oracle estimator in a mean squared error sense.

We implement our estimator using the data of Card (1993). We compare conventional two-stage least squares with our semiparametric feasible reordered IV estimator. While the conventional two-stage least squares estimates are stable across specifications and in some cases statistically significant, the feasible reordered IV estimates are unstable and never significant.

1.1 Related Literature

This paper builds closely on Słoczyński (2026), which introduces weak monotonicity and proposes the reordered IV estimator. More generally, this paper contributes to the literature on the estimation of average treatment effects using instrumental variables. Imbens and Angrist (1994) introduce the monotonicity assumption and show the weighted average representation of the asymptotic two-stage least squares estimand. Angrist and Imbens (1995) extend this result from binary to multivalued treatments, and introduce the saturate and control specification for the case with covariates. Vytlacil (2002) demonstrates that the identifying assumptions of Imbens and Angrist (1994) imply and are implied by a generalised Roy selection model. Building on this, Heckman and Vytlacil (2001, 2005) show how instrumental variables can identify quantities other than the average treatment effect. Kolesár (2013) compares instrumental variable estimators in the context of the heterogeneous treatment effect model with covariates. Kolesár (2013) derives conditions under which the two-stage least squares estimated corresponds to a weighted average of treatment effects, and shows that other estimators say as limited information maximum likelihood may fail to estimate a weighted average of treatment effects. More recently, Blandhol et al. (2022) show that the rich covariates assumption of Kolesár (2013) is unlikely to hold in practice and advocate using more flexible estimators in place of TSLS, such as the double debiased estimators proposed in Chernozhukov et al. (2018) and Singh and Sun (2024).

A second related strand of the literature focuses on variations of the Imbens and Angrist (1994) monotonicity assumption. This includes Klein (2010), de Chaisemartin (2017), Small et al. (2017), and Dahl et al. (2023). Klein (2010) considers the effect of random departures from monotonicity wherein the relationship between the instrument and treatment is subject to independent idiosyncratic perturbations, and characterises the resulting bias in estimates of treatment effects. de Chaisemartin (2017) shows that treatment effects are still identified in the presence of defiers as long as the distribution of compliers over treatment effects nests the distribution of defiers. Small et al. (2017) proposes the stochastic monotonicity assumption, where responses to the instrument are idiosyncratic, but units have unobservable types and monotonicity holds on average for each type. Dahl et al. (2023) assumes that compliers and defiers are both present, but only one type is

present at each point in the potential outcome distribution. This is analogous to weak monotonicity, with switcher type distinguished by potential outcomes rather than covariates. Although potential outcomes are not directly observable, under the assumptions of Dahl et al. (2023) switcher type can be identified from the joint distribution of outcomes and treatments conditional on the instrument. Dahl et al. (2023) estimate treatment effects by direct integration of nonparametric estimates of these distributions.

Outside of the instrumental variable literature, this paper is closely related to Semenova (2025). This paper considers the Lee (2009) bounds for average treatment effects when treatment affects sample participation as well as outcomes. To obtain these bounds, Lee (2009) adopts a monotonicity assumption: treatment either encourages all units to participate in the sample or discourages all units. Semenova (2025) generalises this assumption by allowing treatment to encourage participation for some covariate strata and discourage participation for others. An analogous assumption to weak monotonicity. Semenova (2025) proposes an estimator that combines debiased estimates of bounds for different covariate strata with an estimate of switcher type. The key difference between Semenova (2025) and this paper is the target. The Lee (2009) bounds related to the average treatment effect for always-takers: those who would participate in the sample irrespective of their assigned treatment. In contrast, we target the average treatment effect for switchers. The change in target leads to an estimator with different asymptotic properties.

The remainder of the paper proceeds as follows. Section 2 introduces the reordered IV estimator, and provides a simple example to illustrate. Section 3 develops the general estimator with a semiparametric plug-in. Section 4 presents a simulation exercise, Section 5 contains an application to the data of Card (1993), Section 6 briefly concludes.

Notation: We use capital letters to denote random quantities, and lower case letters to denote fixed values: $X = x$ corresponds to the event that random variable X takes value x. $|\cdot|$ and $||\cdot||$ denote the absolute value and Euclidian norms. A T superscript signifies the transpose.

2 Negative Weights and Reordered IV

This section is divided into four parts. The first summarises known results regarding compliers and defiers. The second introduces the reordered IV estimator. The third sketches its asymptotic distribution with a generic propensity score estimator. The fourth presents a simple example to illustrate the properties of reordered IV.

2.1 Switchers, Compliers, and Defiers

Let $Y \in \mathcal{Y}$ be a scalar outcome variable, $D \in \{0,1\}$ be a scalar binary treatment variable, $Z \in \{0,1\}$ be a scalar binary instrumental variable, and $X \in \mathcal{X} \subset \mathbb{R}^{d_x}$ be a length $d_x \geq 1$ vector of covariates including a constant. $D(z)$ denotes a potential treatment choice given $Z = z$. That is, $D(z)$ is the treatment choice that would be observed if a unit were exogenously assigned the instrument value $Z = z$. Similarly, $Y(d, z)$ denotes a potential outcome given $D = d$ and $Z = z$. We adopt the Imbens and Angrist (1994) identifying assumptions.

Assumption 1 ***Imbens and Angrist (1994) Identifying Assumptions***

(i) Instrument exclusion: $\Pr(Y(d,1) = Y(d,0)|X = x) = 1$ *for any* x.

(ii) Random assignment: $(Y(1), Y(0), D(1), D(0)) \perp\!\!\!\perp Z|X$.

(iii) Monotonicity: Given $X = x$, *either* $D(1) \geq D(0)$ *or* $D(0) \geq D(1)$ *with probability one.*

We define 'switchers'[1] to be units whose potential treatment choice is affected by the instrumental variable in the sense that $D(1) \neq D(0)$. All switchers are either compliers or defiers. For compliers, $D(1) > D(0)$. For defiers, $D(0) > D(1)$. Any unit that is not a switcher is either an always taker ($D(0) = D(1) = 1$) or a never taker ($D(0) = D(1) = 0$).

Under Assumption 1, the proportion of switchers and their type (compliers or defiers) for each X is identified. Let $p^*(x, z)$ be the propensity score

$$p^*(x,z) = \Pr(D = 1|X = x, Z = z) = E[D|X = x, Z = z]. \tag{1}$$

Following Imbens and Angrist (1994),

$$p^*(x,1) - p^*(x,0) = E[D(1)|X = x] - E[D(0)|X = x] = s^*(X) \Pr(D(1) \neq D(0)|X = x), \tag{2}$$

where $s^*(X)$ is

$$s^*(X) = \begin{cases} 1 & p^*(x,1) - p^*(x,0) \geq 0, \\ -1 & p^*(x,1) - p^*(x,0) > 0. \end{cases}$$

Assumption 1 (iii) implies that if $\Pr(D(1) \geq D(0)|X = x) > 0$ then $\Pr(D(0) > D(1)|X = x) = 0$, and vice versa. Hence, $s^*(X)$ delineates whether the switchers for the covariate stratum $X = x$ are compliers or defiers. $\Pr(D(1) \neq D(0)|X)$ is the magnitude of $p^*(X,1) - p^*(X,0)$, and corresponds to the proportion of switchers.

[1] I am grateful to Sami Stouli for suggesting this terminology.

Vytlacil (2002) shows that the Imbens and Angrist (1994) identifying assumptions imply that a selection model of the following form holds for $X = x$

$$D = \mathbb{1}\{\alpha_x + \delta_x Z \geq V_x\} \qquad\qquad V_x \perp\!\!\!\perp Z | X = x, \tag{3}$$

α_x and δ_x are constants common to all units with $X = x$, and V_x is a unit specific random variable. It follows that the propensity score can be expressed as $p^*(x, z) = F_{V_x}(\alpha_x + \delta_x z)$, and $s^*(X) = 2\mathbb{1}\{\delta_x > 0\} - 1$.

In the context of the selection model (3), $\alpha_x + \delta_x z$ is the payoff from taking treatment, and V_x as a unit specific treatment cost. Switchers at $X = x$ are compliers if the instrument increases their net payoff from treatment, and they are defiers if it decreases their net payoff. Both compliers and defiers are present if there exist covariate strata where the instrument makes treatment more attractive, and covariate strata where it makes treatment less attractive.

An alternative interpretation follows from de Chaisemartin (2017). Suppose there are many underlying types, with each type characterised by their treatment effect. de Chaisemartin (2017) shows that Assumption 1 (iii) can be weakened to the assumption that either there are more compliers than defiers of each type, or there are more defiers than compliers. In this case, there are compliers and defiers for all values of X, and $p^*(x, 1) - p^*(x, 0)$ measures the net mass of compliers. Variation in $s^*(X)$ implies that there are some values of x with more compliers than defiers, and some with more defiers than compliers.

2.2 Reordered IV

Consider the two-stage least squares specification

$$\begin{aligned} D &= Z\delta + X^T\alpha + \varepsilon, \\ Y &= D\tau + X^T\beta + u. \end{aligned} \tag{4}$$

Given this specification, the population estimand τ_{tsls} may be written as

$$\tau_{tsls} = \frac{E[Y(Z - L[Z|X = x])]}{E[D(Z - L[Z|X = x])]}, \tag{5}$$

where $L[Z|X]$ denotes the population linear projection of Z on X.[2] Following Kolesár (2013), we impose the restriction that $L[Z|X]$ coincides with the conditional expectation of Z. Blandhol et al. (2022) refer to this assumption as 'rich covariates'.

Assumption 2 (Rich Covariates) $L[Z|X = x] = E[Z|X = x]$.

[2]Specifically, $L[Z|X] = X^T\gamma$ where $\gamma = \text{argmin}_\gamma E[(Z - X^T\gamma)^2]$.

Słoczyński (2026) shows that, under Assumptions 1 and 2, τ_{tsls} may be written as

$$\tau_{tsls} = \frac{E[s^*(X)\omega(x)\tau(x)]}{E[s^*(X)\omega(x)]}, \tag{6}$$

with

$$\begin{aligned} \omega(x) &= \text{Var}(Z|X=x)\ |p^*(x,1) - p^*(x,0)|, \\ \tau(x) &= E[Y(1) - Y(0)|D(1,x) \neq D(0,x), X = x]. \end{aligned} \tag{7}$$

For completeness, Appendix A.1 provides a derivation. $\tau(x)$ is the conditional local average treatment effect (LATE) for units with $X = x$. That is, an average of the treatment effects experienced by the switchers in the $X = x$ stratum. $\omega(x)$ is the proportion of switchers with $X = x$ scaled by the conditional variance of Z.

The decomposition (6) is a weighted average of $\tau(x)$ with the weights $s^*(X)\omega(x)/E[s^*(X)\omega(x)]$. If there are compliers and defiers, then $s^*(X)$ takes values -1 and 1, and some weights are negative. Negative weights are problematic in two ways. First, even if the sign of $\tau(x)$ is positive for all possible values of x, the sign of τ_{tsls} may be negative, and vice versa.[3] Second, notice that the denominator of (6) can be expressed as $\Pr(s^*(X) = 1)E[\omega(x)|s^*(X) = 1] - \Pr(s^*(X) = -1)E[\omega(x)|s^*(X) = -1]$. Its magnitude is therefore determined by the difference in weights between the two types of switchers. If both types are present the denominator becomes smaller. In a sense, the instrument becomes weaker.

The root of the problem is misspecification. The specification (4) imposes that Z and D are monotonically related. In doing so, it gets the sign 'wrong' for one type of switcher. This problem can be resolved by a simple correction to the estimand (5). Suppose we replace $Z - L[Z|X]$ with $s^*(X)(Z - L[Z|X])$. Under Assumptions 1 and 2, the modified estimand can be written as

$$\tau_r = \frac{E[Ys^*(X)(Z - E[Z|X=x])]}{E[Ds^*(X)(Z - E[Z|X=x])]} = \frac{E[\omega(x)\tau(x)]}{E[\omega(x)]}, \tag{8}$$

with all quantities defined as in (7). For completeness, a derivation is given in Appendix A.2. The estimand (8) is guaranteed to be a weighted average with all weights positive.

This correction has a straightforward interpretation. Defiers for the instrument Z are compliers for the instrument $-Z$. It follows that, by construction, all switchers for the transformed instrument $s^*(X)(Z - L[Z|X])$ are compliers, and hence negative weights are no longer an issue.

This reordered IV estimator was originally proposed in Słoczyński (2026), who suggests that the moment condition (8) can be obtained by two-stage least squares in the asymptotic setting with the instrument

$$Z_R = s^*(X)Z + \frac{1 - s^*(X)}{2}.$$

[3] In the terminology of Blandhol et al. (2022), the TSLS estimand is not weakly causal.

Unfortunately, deriving the moment condition is not straightforward. The approach of Słoczyński (2026) leads to

$$\frac{E[Y(Z_R - L[Z_R|X = x])]}{E[D(Z_R - L[Z_R|X = x])]}. \tag{9}$$

Recall that Assumption 2 requires that the conditional expectation of the instrument given X coincides with the linear projection of the instrument on X. Even if we have an instrument that satisfies $E[Z|X = x] = L[Z|X = x]$, it does not follow that $E[Z_R|X = x] = L[Z_R|X = x]$. Suppose that Z and X are independent and that $E[Z] = p_z \neq 1/2$. As long as X includes a constant, $E[Z|X = x] = L[Z|X = x]$ is immediate. However,

$$E[Z_R|X = x] = s^*(x)p_z + \frac{1 - s^*(x)}{2},$$

which is not a linear function of X. Hence, $E[Z_R|X = x] \neq L[Z_R|X = x]$. The moment condition (8) can, however, be obtained if we consider two-stage least squares in the asymptotic setting with the transformed instrument $s^*(x)(Z - L[Z|X = x])$. Under Assumption 2,

$$E[s^*(X)(Z-L[Z|X])|X = x] = s^*(x)(E[Z|X = x]-L[Z|X = x]) = s^*(x)(E[Z|X = x]-E[Z|X = x]) = 0,$$

and hence $E[s^*(X)(Z - L[Z|X)|X = x] = L[s^*(X)(Z - L[Z|X])|X = x]$. Notice that, for this result to hold, it is necessary to orthogonalise the instrument before reordering. The conditional expectation of the transformed instrument $s^*(X)Z$ also fails to coincide with its linear projection on X.

2.3 The Asymptotic Distribution of Reordered IV

The discussion above assumes that $s^*(X)$ is known, whereas in general it must be estimated from the data. Here we consider a feasible reordered IV estimator, with $s^*(X)$ constructed from propensity score estimates. Equation (2) implies that

$$s^*(X) = 2\mathbb{1}\{p^*(X, 1) - p^*(X, 0) \geq 0\} - 1.$$

Given an estimated propensity score function $\hat{p}(X, Z)$, we can define an estimated sign function $\hat{s}(X)$ as

$$\hat{s}(X) = 2\mathbb{1}\{\hat{p}(X, 1) - \hat{p}(X, 0) \geq 0\} - 1. \tag{10}$$

A feasible reordered IV estimator based on the sample analog of the moment condition (8) is then

$$\hat{\tau}_r = \frac{\sum_{i=1}^n \hat{s}(X_i)Y_i(Z_i - X_i^T\hat{\gamma})/n}{\sum_{i=1}^n \hat{s}(X_i)D_i(Z_i - X_i^T\hat{\gamma})/n}, \tag{11}$$

where $\hat{s}(X)$ is as defined in (10). $X^T\hat{\gamma}$ is the sample linear projection of Z on X. Under Assumption 2 this is a consistent estimator for $E[Z|X = x]$. To understand the asymptotics of the feasible estimator (11), it is helpful to define the following oracle estimator

$$\hat{\tau}_o = \frac{\sum_{i=1}^n s^*(X_i)Y_i(Z_i - X_i^T\hat{\gamma})/n}{\sum_{i=1}^n s^*(X_i)D_i(Z_i - X_i^T\hat{\gamma})/n}. \tag{12}$$

This oracle is still subject to finite sample noise and must estimate γ, but has full knowledge of $s^*(X)$.

The following assumptions place restrictions on the behaviour of the estimator $\hat{p}(x, z)$ and the distribution of $p^*(X, 1) - p^*(X, 0)$. Some additional regularity conditions are stated in Assumption 8 in Appendix B.

Assumption 3 ($\sqrt{n}$-Uniform Consistency) *Let* $||\hat{p} - p^*||_\infty = \sup_{x\in\mathcal{X}, z\in\{0,1\}} |\hat{p}(x, z) - p^*(x, z)|$,

$$||\hat{p} - p^*||_\infty = O_p(n^{-1/2})$$

Assumption 4 (VC-Class) *Define the class of functions*

$$\mathcal{S} \coloneqq \{s(X) = 2\mathbb{1}\{p(X, 1) \geq p(X, 0)\} - 1 : p \in \mathcal{P}\}$$

where $\mathcal{P}$ is the class of propensity score functions. $\mathcal{S}$ is a Vapnik-Chervonenkis (VC) subgraph class of functions.

Assumption 5 (Margin Condition) *There exist $C > 0$, $\eta > 0$, and $\zeta > 0$ such that for $0 < t \leq \eta$,*

$$\Pr(|p^*(x, 1) - p^*(x, 0)| \leq t) \leq Ct^\zeta.$$

Assumption 3 imposes that the estimator of the propensity score is uniformly consistent. We impose this form of consistency as it enables results to be derived in a transparent manner. Assumption 4 restricts the complexity of the decision boundary induced by the propensity score functions. It rules out decision boundaries that are arbitrarily flexible.[4] This Assumption is used to deal with several empirical process type terms. Assumptions 3 and 4 hold for a variety of functional forms. For example, if covariates have bounded support, they hold for any parameteric single index model, such as probit and logit, where the index function is Lipschitz continuous. This is discussed in more detail in Section 3. Assumption 5 rules out distributions of $|p^*(x, 1) - p^*(x, 0)|$ that have mass points at 0, and densities that grow too rapidly in this neighbourhood. If $|p^*(x, 1) - p^*(x, 0)|$ has a bounded density, then Assumption 5 holds with $\zeta = 1$. Margin conditions are also used in Kitagawa and Tetenov (2018) and Semenova (2025).

[4] In other words, the decision boundary cannot be too 'wiggly'.

The following theorem establishes the asymptotic distributions of the feasible and oracle estimators

Theorem 1 *Suppose Assumptions 1, 2 and 8 hold, then*

$$\sqrt{n}(\hat{\tau}_o - \tau_r^*) \xrightarrow{d} N(0, \sigma_{\tau_o})$$

If, in addition, Assumptions 3 - 5 hold, then

$$\sqrt{n}(\hat{\tau}_r - \hat{\tau}_o) = o_p(1)$$

where τ_r^* is the asymptotic estimand given in (8). A proof is given in Appendix B. This theorem states that the oracle estimator is $\sqrt{n}$-consistent and asymptotically normal, and that the feasible estimator converges in probability to the oracle estimator. That is, estimating $s^*(X)$ does not affect the asymptotic distribution of the estimator. The theorem below provides an estimator for the asymptotic variance $\sigma_{\tau_o}^2$

Theorem 2 *Write* $\hat{e}_{Zi} = Z_i - X_i^T\hat{\gamma}$ *and* $\hat{e}_{Yi} = Y_i - \hat{\tau}_r D_i$. *Let*

$$\hat{\sigma}_{\tau_o}^2 = \frac{\sum_{i=1}^n \hat{e}_{Zi}^2(\hat{e}_{Yi}\hat{s}(X_i) - (\sum_{i=1}^n \hat{e}_{Yi}\hat{s}(X_i)X_i)^T(\sum_{i=1}^n X_i X_i^T/n)^{-1}X_i)^2/n}{(\sum_{i=1}^n D_i\hat{s}(X_i)\hat{e}_{Zi}/n)^2}. \tag{13}$$

Suppose Assumptions 1 - 5, 8, and 9 hold. Then

$$\hat{\sigma}_{\tau_o}^2 - \sigma_{\tau_o}^2 = o_p(1)$$

These standard errors are similar to those obtained in White (1982).

standard errors do not assume homoskedasticity.

The result of Theorem 1 can be understood as follows. The feasible and oracle estimators differ only in replacing $s^*(X)$ with the estimate $\hat{s}(X)$. In a decomposition, replacing $s^*(X)$ with $\hat{s}(X)$ leads to additional terms like $E[(\hat{s}(X) - s^*(X))D(Z - X^T\gamma^*)]$, which adjust for the plug-in estimate. These terms are bounded in magnitude by the expectation of $\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}$ multiplied by $|p^*(X,1) - p^*(X,0)|$. For example,

$$\begin{aligned}|E[(\hat{s}(X) - s^*(X))D(Z - X^T\gamma^*)]| &= |E[(\hat{s}(x) - s^*(x))\mathrm{Var}(Z|X=x)(p^*(x,1) - p^*(x,0)]|, \\ &\leq CE[\mathbb{1}\{\hat{s}(x) \neq s^*(x)\}|p^*(x,1) - p^*(x,0)|].\end{aligned}$$

Under Assumption 3, $|p^*(x,1) - p^*(x,0)|$ is small whenever $\mathbb{1}\{\hat{s}(X) \neq s^*(X)\} = 1$. The event $\hat{s}(x) \neq s^*(x)$ indicates a classification error. If $\hat{s}(X) = -1$ and $s^*(X) = 1$, compliers are misclassified as defiers. If $\hat{s}(X) = 1$ and $s^*(X) = -1$, defiers are misclassified as compliers. Using the form of the sign function,

$$\hat{s}(x) \neq s^*(x) \implies |p^*(x,1) - p^*(x,0)| \leq |\hat{p}(x,1) - \hat{p}(x,0) - (p^*(x,1) - p^*(x,0))|.$$

The decision boundary for $s^*(x)$ is $p^*(x,1)-p^*(x,0)=0$. $|p^*(x,1)-p^*(x,0)|$ measures the distance from this boundary. Instead of $p^*(x,z)$, $\hat{s}(X)$ uses the estimated propensity score function $\hat{p}(x,z)$ to classify switchers. For missclassification to occur, the distance between $\hat{p}(x,1)-\hat{p}(x,0)$ and $p^*(x,1)-p^*(x,0)$, must exceed $|p^*(x,1)-p^*(x,0)|$. The sup norm $||\hat{p}-p^*||_\infty$ provides a bound on this distance,

$$|\hat{p}(x,1)-\hat{p}(x,0)-(p^*(x,1)-p^*(x,0))| \leq 2||\hat{p}-p^*||_\infty.$$

Hence,

$$\begin{aligned}
E[\mathbb{1}\{\hat{s}(x)\neq s^*(x)\}|p^*(x,1)-p^*(x,0)|] &\leq E[[\mathbb{1}\{|p^*(x,1)-p^*(x,0)|\leq 2||\hat{p}-p^*||_\infty\}|p^*(x,1)-p^*(x,0)|], \\
&\leq ||\hat{p}-p^*||_\infty \Pr(|p^*(x,1)-p^*(x,0)|\leq 2||\hat{p}-p^*||_\infty), \\
&\leq C||\hat{p}-p^*||_\infty^{1+\zeta},
\end{aligned}$$

where the final line uses Assumption 5. By Assumption 3, $||\hat{p}-p^*||_\infty^{1+\zeta}$ is $O_p(n^{-(1+\zeta)/2})$, so $\sqrt{n}||\hat{p}-p^*||_\infty^{1+\zeta}$ is $o_p(1)$, and these terms vanish asymptotically.

Intuitively, the correct classification of switchers for $X=x$ does not require that the propensity score function is estimated perfectly, merely that that estimate is within a margin $|p^*(x,1)-p^*(x,0)|$. This margin is directly related to the weights in (11). Where weights are large, switchers are easy to classify as the margin $|p^*(x,1)-p^*(x,0)|$ is large. Asymptotically, classification mistakes occur only where weights are small. This means that their effect on the estimand is negligible.

2.4 A Simple Example

Suppose that covariates consist of a constant and a single categorical variable X, and that Z is independent of X

$$\Pr(Z=1)=\Pr(Z=0)=1/2.$$

Let X take values $\{-K,...,-1,1,...,K\}$. We assume that X is fixed in repeated samples: a sample of size n is comprised of $n_x=n/2K$ observations for each value of X. Consider the selection model

$$D=\mathbb{1}\{\delta_x Z\geq V\},$$

with δ_x chosen so that $p^*(x,1)-p^*(x,0)=\kappa x$, where $\kappa>0$. For $X<0$, switchers are defiers, for $X>0$ switchers are compliers, and the mass of switchers increases linearly with the magnitude of X. As the propensity score distribution has no mass at $p^*(x,1)=p^*(x,0)$, Assumption 5 holds. Notice that $E[D(Z-E[Z|X])]=0$, so the standard two-stage least squares estimand does not

exist. The outcome process is

$$\begin{aligned} Y &= Y(0) + D(Y(1) - Y(0)), \\ Y(0) &= U(0), \\ Y(1) &= \beta + U(1), \end{aligned}$$

where $U(0)$ and $U(1)$ are potential outcome residuals. We assume that $(U(1), U(0), V) \perp\!\!\!\perp (X, Z)$ and $(U(1), U(0), V) \sim N(0, \Sigma)$, where Σ is a covariance matrix. For simplicity, the diagonal elements of Σ are set to 1. In the case of V, this is without loss of generality.

In this setting, the quantities defined in (7) can be explicitly calculated. For $X = x$,

$$\omega(x) = \frac{1}{4}\kappa|x| \qquad s^*(x) = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \end{cases} \qquad \tau(x) = \beta + (\rho_{1V} - \rho_{0V}) s^*(X) \frac{\phi(0) - \phi(\delta_x)}{\kappa|x|},$$

where ρ_{1V} is the correlation between $U(1)$ and V, ρ_{0V} is correlation between $U(0)$ and V, and $\phi(\cdot)$ is the standard normal density. Using the decomposition (8), the asymptotic reordered IV estimand is

$$\frac{\sum_x \Pr(X = x)\omega(x)\tau(x)}{\sum_x \Pr(X = x)\omega(x)} = \beta + \frac{(\rho_{1V} - \rho_{0V}) \sum_x s^*(X)(\phi(0) - \phi(\delta_x))}{\sum_x \kappa|x|} = \beta,$$

where the symmetry of the δ_x results in $\sum_x s^*(X)(\phi(0) - \phi(\delta_x)) = 0$.

Given this data generating process, we implement both the feasible and oracle reordered IV estimators. We consider a saturate and control type specification. Estimates of $E[Z|X]$ are obtained by OLS estimation of the specification

$$Z = \sum_x \mathbb{1}\{X = x\} g_x + e_Z. \tag{14}$$

In this case $\hat{E}[Z|X = x] = \hat{g}_x$, and $\hat{g}_x$ is the sample conditional mean of Z given $X = x$. Notice that Assumption 2 is satisfied in this case. The oracle estimate is calculated by (12) with the estimate $\hat{g}_x$ plugged-in.

To calculate the feasible estimator, we also require $\hat{s}(X)$. Estimates of the propensity score are obtained by OLS estimation of

$$D = \sum_x \mathbb{1}\{X = x\}(a_x + d_x Z) + e_D. \tag{15}$$

Given this specification, $\hat{p}(x, z) = \hat{a}_x + \hat{d}_x z$, a_x is the sample mean of D conditional on $X = x, Z = 0$, and d_x is the difference between the sample mean of D conditional on $X = x, Z = 1$ and $X = x, Z = 0$. It is easy to show that this estimator satisfies Assumption 3. With this functional

	$\kappa\|x\| = 0.05$	$\kappa\|x\| = 0.1$	$\kappa\|x\| = 0.15$	$\kappa\|x\| = 0.2$	Overall
$n_x = 50$	0.363	0.240	0.143	0.074	0.853
$n_x = 100$	0.309	0.158	0.064	0.020	0.715
$n_x = 200$	0.240	0.078	0.016	0.002	0.525
$n_x = 400$	0.158	0.022	0.001	0.000	0.324

Table 1: Probability of misclassification given n_x.

form, $\hat{s}(x) = \mathbb{1}\{\hat{d}_x \geq 0\}$.[5] The feasible estimate is calculated by (11) with $\hat{g}_x$ and $\hat{s}(X)$ plugged in. In addition, estimates of the standard error are calculated by (13), also with $\hat{g}_x$ and $\hat{s}(X)$ plugged in.

We set $K = 4$, $\kappa = 0.05$. This implies that switcher shares are between 0.05 and 0.2. The number of observations for $X = x$ is $n_x = 50$, 100, 200, and 400, with the total sample size given by $8n_x$. The results of Section 2.3 imply that the sampling distribution of the feasible estimator should converge to that of the oracle estimator as misclassification becomes concentrated at covariate values with few switchers. Table 1 shows misclassification probabilities calculated by taking the expectation of $\mathbb{1}\{\hat{s}(x) \neq s^*(x)\}$ using the conditional joint distribution of (D, Z). The first four columns show misclassification probabilities for a given switcher share $\kappa|x|$. The final column is the probability that misclassification occurs for at least one value of X. Even with $n_x = 400$, misclassification occurs in a little under 1/3 of draws. For simplicity, β is set to 0, so the asymptotic reordered IV estimand is zero. The off-diagonal elements of Σ are $\rho_{1V} = -0.5$, $\rho_{0V} = 0.5$ and $\rho_{01} = -0.25$ (the correlation between U_0 and U_1). The value of Σ is irrelevant asymptotically, but may matter in finite samples. As V is resistance to treatment, these correlations imply a positive selection effect: those who opt for treatment will have, on average, more positive treatment effects.

Table 2 contains summary statistics for $\hat{\tau}_r$ and $\hat{\tau}_o$ calculated from 1,000 draws for each sample size. Figure 1 plots the densities of $\sqrt{n}(\hat{\tau}_r - \tau_r)$ and $\sqrt{n}(\hat{\tau}_0 - \tau_r)$. The sampling distributions behave in line with the results of Section 2.3. For $n_x = 50$, the two are disimilar, but converge as the sample size increases. The sampling distribution of the oracle estimator is more dispersed than that of the feasible estimator, particularly in the smaller sample sizes. This arises due to negative weights. The oracle estimand is a weighted sum with nonnegative weights when the data is distributed according to the population distribution. The feasible estimator used here is, by construction, a weighted sum with nonngeative weights for any sample.[6] When $s^*(X) \neq \hat{s}(X)$,

[5] If all observations for $X = x$ have the same value of Z, $\hat{s}(x)$ is set to 0.

[6] The estimated sign function specified here can be written as

$$\hat{s}(x) = \text{sign}\left(\frac{\sum_i \mathbb{1}\{X_i = x\} Z_i D_i}{\sum_i \mathbb{1}\{X_i = x\} Z_i} - \frac{\sum_i \mathbb{1}\{X_i = x\}(1 - Z_i) D_i}{\sum_i \mathbb{1}\{X_i = x\}(1 - Z_i)}\right).$$

the oracle estimator includes negative weights. This decreases the magnitude of the denominator, which increases the variance of the estimator. In terms of mean squared error (MSE), the feasible estimator is preferable to the oracle estimator, but the oracle estimator generally exhibits slightly less bias. Table 3 presents coverage rates for confidence intervals calculated using the estimated standard error defined in (13) and the feasible and oracle estimates. For smaller sample sizes, the confidence intervals exhibit slight under coverage for the oracle estimates and slight over coverage for the feasible estimates, but both dimish as the sample size increases. This is consistent with the results in Table 2. The finite sample oracle estimator is noisier than its asymptotic counterpart, whereas the feasible estimator is more stable than its asymptotic counterpart. As a consequence, the asymptotic standard error is too small for the oracle estimator and too large for the feasible estimator.

Given the results of Section 3, we would expect the feasible reordered IV estimator should converge to the oracle estimator, as is the case here. However, the specific form this convergence takes here, the feasible estimator being more stable than the oracle estimator, is a function of the saturate and control specification. In particular, it rests on $\hat{s}(X)$ being equal to the sign of the sample conditional covariance for each X, which is not an implication of Assumption 3 and does not hold for a generic propensity score estimator.

3 Semiparametric Reordered IV

The example in Section 2.4 uses a saturate and control-like specification with $s^*(X)$ estimated as the difference in sample conditional means. This specification is only feasible when there are no continuous covariates, and the number of covariate strata is small relative to the sample size. In this section, we develop a general reordered IV estimator which is not subject to these limitations.

3.1 Propensity Score Functional Form

To construct a general estimator, we impose a semiparametric functional form on the propensity score function.

Let $\bar{Z}(x) = \sum_i \mathbb{1}\{X_i = x\}Z_i / \sum_i \mathbb{1}\{X_i = x\}$. Given that $E[Z|X]$ is estimated by $\bar{Z}(x)$, for a generic sign function $s(X)$, the denominator of the reordered IV estimator is

$$\frac{n_x}{n} \sum_x s^*(X)\bar{Z}(x)\left(1 - \bar{Z}(x)\right)\left(\frac{\sum_i \mathbb{1}\{X_i = x\}Z_i D_i}{\sum_i \mathbb{1}\{X_i = x\}Z_i} - \frac{\sum_i \mathbb{1}\{X_i = x\}(1 - Z_i)D_i}{\sum_i \mathbb{1}\{X_i = x\}(1 - Z_i)}\right).$$

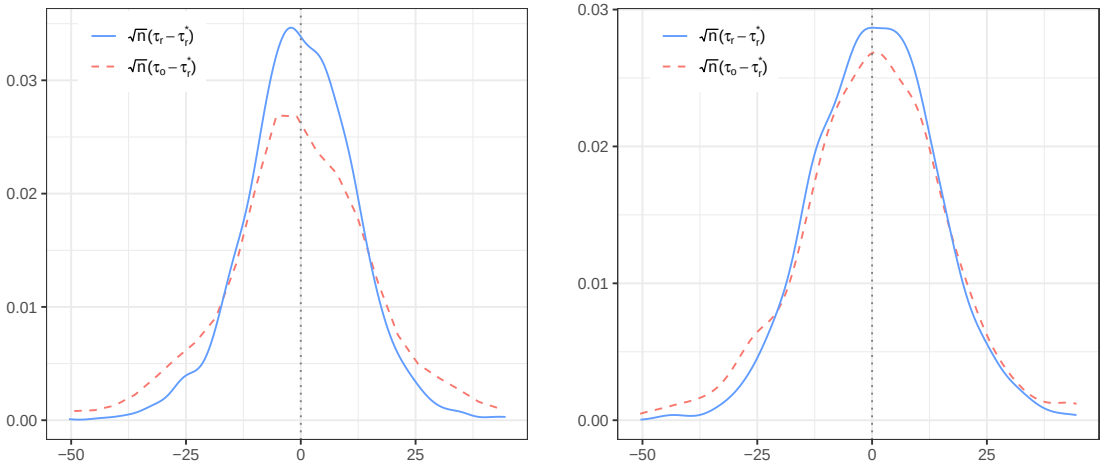
If $s^*(X)$ coincides with $\hat{s}(x)$, then the denominator is a sum of positive terms. If $s^*(X) \neq \hat{s}(x)$ for some x, then the denominator includes negative terms.

n_x	Mean	Standard Error	MSE	Median	First Quartile	Third Quartile
Oracle Estimator						
50	-0.030	3.305	10.924	-0.048	-0.473	0.492
100	0.001	0.584	0.341	0.021	-0.324	0.360
200	-0.012	0.399	0.159	-0.002	-0.270	0.238
400	-0.016	0.281	0.079	-0.015	-0.192	0.170
Feasible Estimator						
50	0.009	0.620	0.384	-0.020	-0.365	0.389
100	0.029	0.478	0.230	0.029	-0.288	0.335
200	-0.013	0.373	0.140	-0.002	-0.251	0.229
400	-0.009	0.267	0.072	-0.014	-0.185	0.162

Table 2: Summary statistics for the sampling distribution of the oracle and feasible estimators. Results are for 1,000 draws.

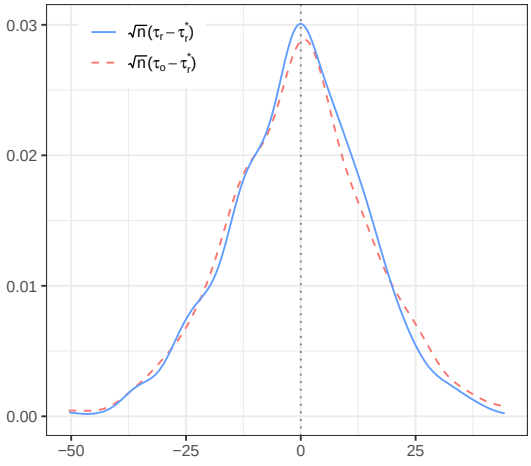
	$\hat{\tau}_r$			$\hat{\tau}_o$		
n_x	90%	95%	99%	90%	95%	99%
50	0.962	0.992	0.999	0.801	0.859	0.904
100	0.933	0.981	0.998	0.870	0.933	0.970
200	0.913	0.967	0.988	0.882	0.946	0.978
400	0.899	0.955	0.989	0.881	0.939	0.976

Table 3: Coverage rates of the 90%, 95% and 99% confidence intervals. Results are for 1,000 draws.

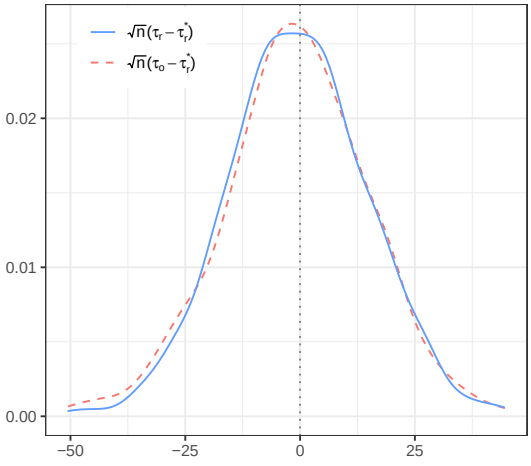


(a) $n_x = 50, n = 400$

(b) $n_x = 100, n = 800$



(c) $n_x = 200, n = 1600$



(d) $n_x = 400, n = 3200$

Figure 1: Densities of $\sqrt{n}(\hat{\tau}_r - \tau_r^*)$ and $\sqrt{n}(\hat{\tau}_o - \tau_r^*)$. n is the sample size, n_x is the number of observations for rach value of X. Densities are calculated using results from 1,000 draws and the rule-of-thumb bandwidth.

Assumption 6 (Functional Form) *The propensity score may be written as*

$$p^*(X, Z) = \varphi(X^T\alpha + ZX^T\delta)$$

where $\varphi(\cdot) : \mathbb{R} \to [0, 1]$ *is an unknown strictly increasing function, and* $\alpha \in \mathcal{A} \subset \mathbb{R}^{d_x}$ *and* $\delta \in \mathcal{D} \subset \mathbb{R}^{d_x}$ *are vectors of parameters.*

Assumption 6 states that the propensity score function is a single index model with an unknown monotonic link function $\varphi(\cdot)$. This functional form accommodates all parametric single index models, such as probit and logit, but is more flexible as the link function is left unspecified.

Assumption 6 leads to a simple form for $s^*(X)$. Recall that $s^*(X) = 2\mathbb{1}\{p^*(x, 1) \geq p^*(x, 0)\} - 1$. Imposing $p^*(x, z) = \varphi(x^T\alpha + zx^T\delta)$ and using that $\varphi(\cdot)$ is strictly increasing,

$$p^*(x, 1) \geq p^*(x, 0) \iff x^T\delta \geq 0. \tag{16}$$

It follows that, to construct an estimate of $s^*(X)$, we require only estimates of the interaction parameters δ. The function $\varphi(\cdot)$ is irrelevant. This is advantageous as, in semiparametric models, the parametric component is typically estimated at a faster rate than the nonparametric component.

Section 2.3 showed asymptotic normality and oracle equivalence of the feasible reordered IV estimator provided that propensity score estimates satisfy uniform consistency (Assumption 3) and $s^*(X)$ belongs to a VC subgraph class of functions (Assumption 4). As $x^T\delta$ is a finite dimensional vector space, it is straightforward to show that $s^*(X) = \mathbb{1}\{x^T\delta \geq 0\}$ belongs to a VC subgraph class. Thus, Assumption 6 implies Assumption 4. However, Assumption 3 does not follow from Assumption 6. We replace it with Assumption 7, which plays the same role. Let δ^* be the true value of δ and $\hat{\delta}$ be an estimator.

Assumption 7

(i) $\hat{\delta} - \delta^* = O_p(n^{-1/2})$

(ii) There exists a universal constant $C > 0$ *such that, for any* $x \in \mathcal{X}$*,*

$$|p^*(x, 1) - p^*(x, 0)| \leq C|x^T\delta^*|$$

(iii) $E[||X||] < \infty$

Assumption 3 was used to bound (in probability) terms such as $\Pr(\hat{s}(X) \neq s^*(X))$. Assumption 7 fulfills the same role. Given Assumption 6, for fixed $X = x$

$$\hat{s}(x) \neq s^*(X) \implies |x^T\delta^*| \leq |x^T(\delta^* - \hat{\delta})| \leq ||x|| \cdot ||\delta^* - \hat{\delta}||$$

Combining with Assumption 7 (ii), if $\hat{s}(x) \neq s^*(x)$

$$|p^*(x,1) - p^*(x,0)| \leq Cx^T\delta^* \leq C||x|| \cdot ||\delta^* - \hat{\delta}||$$

so that

$$\Pr(\hat{s}(x) \neq s^*(x)) \leq \Pr(|p^*(x,1) - p^*(x,0)| \leq C||x|| \cdot ||\delta^* - \hat{\delta}||)$$

Assumption 7 (iii) implies that $||x||$ is bounded with high probability. Assumption 5, Assumption 7 (i), and a union bound argument imply that $\Pr(\hat{s}(x) \neq s^*(x)) = o_p(1)$.

3.2 Estimation of $s^*(X)$

As we do not require $\varphi(\cdot)$, we can use an estimator which avoids explicitly estimating it: the Maximum Rank Correlation (MRC) estimator of Han (1987).

To introduce this estimator, let $X = (1, X_1)$ where X_1 contains all non-constant conditioning covariates apart from the constant. Write $\alpha = (\alpha_0, \alpha_1)$ and $\delta = (\delta_0, \delta_1)$ so that $X^T\alpha = \alpha_0 + X_1^T\alpha_1$ and $ZX^T\delta = \delta_0 Z + Z(X^T\delta_1)$. Define the normalised parameter vector

$$(\tilde{\alpha}_1, \tilde{\delta}_0, \tilde{\delta}_1) \coloneqq \frac{(\alpha_1, \delta_0, \delta_1)}{||(\alpha_1, \delta_0, \delta_1)||},$$

and say $(\tilde{\alpha}_1, \tilde{\delta}_0, \tilde{\delta}_1) \in \tilde{\mathcal{A}}_1 \times \tilde{\mathcal{D}}_0 \times \tilde{\mathcal{D}}_1$. Let n_1 be the number of observations with $D = 1$, and n_0 the number with $D = 0$. The MRC estimator is

$$(\hat{\tilde{\alpha}}_1, \hat{\tilde{\delta}}_0, \hat{\tilde{\delta}}_1) = \underset{(\tilde{\alpha}_1, \tilde{\delta}_0, \tilde{\delta}_1) \in \tilde{\mathcal{A}}_1 \times \tilde{\mathcal{D}}_0 \times \tilde{\mathcal{D}}_1}{\operatorname{argmax}} \binom{N}{2}^{-1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} \mathbb{1}\{X_{1i}^T\tilde{\alpha}_1 + Z_i\tilde{\delta}_0 + Z_i X_{1i}^T\tilde{\delta}_1 > X_{1j}^T\tilde{\alpha}_1 + Z_j\tilde{\delta}_0 + Z_j X_{1j}^T\tilde{\delta}_1\}. \tag{17}$$

This criterion sums over all pairs of observations i, j such $D_i = 1$ and $D_j = 0$. A pair contributes 1 to the criterion if $X_{1i}^T\tilde{\alpha}_1 + Z_i\tilde{\delta}_0 + Z_i X_{1i}^T\tilde{\delta}_1 > X_{1j}^T\tilde{\alpha}_1 + Z_j\tilde{\delta}_0 + Z_j X_{1j}^T\tilde{\delta}_1$, and 0 otherwise. The total contribution of any observation i with $D_i = 1$ is equal to its rank among all n_0 observations with $D_j = 0$. $(\hat{\tilde{\alpha}}_1, \hat{\tilde{\delta}}_0, \hat{\tilde{\delta}}_1)$ is chosen so that no change in value would increase the average rank of $D_i = 1$ observations among the $D_j = 0$ observations. This estimator does not identify α_0 as adding a constant to the index does not change ranks. Similarly, multiplying all parameters by a positive constant does not change ranks, so parameters are only identified up to scale. To construct $\hat{s}(x)$, we require only the sign of $x^T\delta$, which is unchanged if we multiply δ by a positive constant. Hence, estimating $\tilde{\delta} = (\tilde{\delta}_0, \tilde{\delta}_1)$ is sufficient.

Han (1987) proves that the MRC estimates of $(\tilde{\alpha}_1, \tilde{\delta}_0, \tilde{\delta}_1)$ are consistent, and Sherman (1993) that they are $\sqrt{n}$-consistent and asymptotically normal. Subbotin (2007) shows that standard errors can be estimated by the standard nonparametric bootstrap. The MRC estimator does not explicitly compute the nonparametric function $\varphi(\cdot)$, which simplifies estimation. On the other

hand, the criterion function (17) is neither smooth nor globally concave. Maximisation must be performed numerically by an algorithm that does not use gradient information, and there may be multiple local maxima. Shin and Todorov (2021) show that the maximisation problem can be formulated as a mixed integer programming problem, and the global maximum obtained using a branch and bound algorithm. We use a simplex method with multiple starting points.

3.3 Estimator Algorithm and Asymptotic Distribution

The general feasible reordered IV estimator can be calculated as follows.

1. Obtain estimates $\hat{\tilde{\delta}}$ using the MRC estimator. Calculate the estimated sign function as

$$\hat{s}(X_i) = 2\mathbb{1}\{X_i^T\hat{\tilde{\delta}} \geq 0\} - 1$$

2. Regress Z on X and save the fitted values $X^T\hat{\gamma}$.

3. Calculate $\hat{\tau}_r$ as

$$\hat{\tau}_r = \frac{\sum_{i=1}^n Y_i\hat{s}(X_i)(Z_i - X_i^T\hat{\gamma})}{\sum_{i=1}^n D_i\hat{s}(X_i)(Z_i - X_i^T\hat{\gamma})}.$$

The following Theorem characterises the asymptotic distribution of $\hat{\tau}_r$ with a generic estimator of $\hat{\delta}$. The result is identical to Theorem 1, but Assumptions 3 and 4 are replaced by Assumptions 6 and 7.

Theorem 3 *Suppose Assumptions 1, 2, 5 - 7 and 8 hold. Then*

$$\sqrt{n}(\hat{\tau}_r - \hat{\tau}_o) = o_p(1)$$

In addition, the formula (13) *remains a consistent estimator of* σ_{τ_o}.

A proof is presented in Appendix B. Showing that Theorem 3 applies when the MRC estimator is used requires verifying Assumption 7 (i). Sherman (1993) shows asymptotic normality when X contains at least one variable with everywhere positive density (conditional on the other elements of X). In the case where X contains only discrete variables, the argument of Section 5 of Cavanagh and Sherman (1998) gives a faster than $\sqrt{n}$ rate of convergence.[7] An additional complication is that Assumption 7 (ii) is stated in terms of δ, while the MRC estimator instead estimates scaled parameters $\tilde{\delta}$. However, Assumption 7 (ii) can be restated for the scaled parameters.

[7] This requires that the link function is strictly increasing, which is assumed in Assumption 6.

4 Simulations

In this section, we evaluate the finite sample performance of the estimator proposed in Section 3 through a simulation exercise. Maintaining that $X = (1, X_1)$ with X_1 collecting all non-constant covariates, the data generating process is

$$\begin{aligned} &X_1 \sim N(0, I) \qquad Z \sim \text{binomial}(1/2), \\ &D = \mathbb{1}\{Z(\delta_0 + X_1^T \delta_1) \geq V\}, \\ &Y(0) = U(0), \\ &Y(1) = \beta_0 + X_1^T \beta_1 + U(1), \\ &Y = DY(1) + (1 - D)Y(0), \\ &(V, U(1), U(0)) \sim N(0, \Sigma). \end{aligned} \tag{18}$$

Without loss of generality, the variance of V is normalised to one, but Σ is otherwise unrestricted.

For a particular stratum $(1, x_1)$, the proportion of switchers is $|\Phi(\delta_0 + x_1^T \delta_1) - \Phi(0)|$. The switchers are compliers if $\delta_0 + x_1^T \delta_1 \geq 0$, and defiers otherwise. Integrating over strata, the proportion of compliers in the population is

$$\theta_c = \int_{-\delta_0}^{\infty} (\Phi(\delta_0 + t) - \Phi(0)) \phi\left(\frac{t}{||\delta_1||}\right) dt \tag{19}$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ denote the standard normal distribution and density. The proportion of defiers is

$$\theta_d = \int_{-\infty}^{-\delta_0} (\Phi(0) - \Phi(\delta_0 + t)) \phi\left(\frac{t}{||\delta_1||}\right) dt \tag{20}$$

For any δ_1 such that $||\delta_1|| > 0$, both compliers and defiers exist. Larger values of $||\delta_1||$ increase both the proportions of compliers and defiers. An increase in the value of δ_0 increases the proportion of compliers, and decreases the proportion of defiers.

For this data generating process, the population estimand can be expressed as

$$\tau_r^* = \beta_0 + \frac{\beta_1^T \delta_1}{||\delta_1||^2} \tau_x^* + (E[U(1)V] - E[U(0)V])\tau_u^* \tag{21}$$

with

$$\tau_x^* = \frac{1}{\theta_c + \theta_d} \left(\int_{-\delta_0}^{\infty} (\Phi(\delta_0 + t) - \Phi(0)) t \phi\left(\frac{t}{||\delta_1||}\right) dt + \int_{-\infty}^{-\delta_0} (\Phi(0) - \Phi(\delta_0 + t)) t \phi\left(\frac{t}{||\delta_1||}\right) dt \right)$$

$$\tau_u^* = \frac{1}{\theta_c + \theta_d} \left(\int_{-\delta_0}^{\infty} (\phi(0) - \phi(\delta_0 + t)) \phi\left(\frac{t}{||\delta_1||}\right) dt + \int_{-\infty}^{-\delta_0} (\phi(\delta_0 + t) - \phi(0)) \phi\left(\frac{t}{||\delta_1||}\right) dt \right)$$

τ_r^* can thus be expressed as the sum of the average treatment effect (β_0) and two additional terms. The first reflects heterogeneity in treatment effects due to the observable controls X, the second reflects heterogeneity in treatment effects due to the unobservables $(U(1), U(0), V)$.

	Design 1	Design 2
θ_c	0.11	0.10
θ_d	0.09	0
$\frac{\beta_1^T \delta_1}{\|\|\delta_1\|\|^2} \tau_x^*$	-0.05	0
$(E[U(1)V] - E[U(0)V])\tau_u^*$	-0.05	-0.10

Table 4: Targets

In our simulations, we fix the shares of compliers and defiers, and the population estimand. Equations (19), (20), and (21) imply that are determined by δ_0, $||\delta_1||$, β_0, $\beta_1^T\delta_1$, and $E[U(1)V] - E[U(0)V]$. Hence, we back out values for these parameters, and choose other parameters arbitrarily.

For simplicity, we set $\beta_0 = 0$. For the remaining parameters, we consider two designs. Targets in both cases are given in Table 4. The first is a case where weak monotonicity holds and conventional TSLS would be highly unreliable. Here we fix the population share of compliers at 0.11, the share of defiers at 0.09. The second is a case with only compliers, so strong monotonicity holds, but the instrument is weak. Here the share of compliers is 0.10, and the share of defiers is 0. In both cases, parameters are chosen so that τ_r^* is -0.10. Further details of the parameterisation are presented in Appendix C.

As in Section 2.4, we consider the oracle reordered IV estimator and a feasible reordered IV estimator. Here we use the semiparametric estimator of Section 3 rather than the saturate and control specification of Section 2.4. In both cases, $E[Z|X]$ is estimated as the sample mean of X. As the support X_1 is unbounded and Z is binary, $E[Z|X]$ cannot be a linear function of any element of X_1. Given this estimate, the oracle estimator is then calculated according to (12). For the feasible estimator, we must also construct $\hat{s}(X)$. We specify the propensity score as

$$\varphi(\alpha_0 + X_1^T\alpha_1 + Z(\delta_0 + X_1^T\delta_1)).$$

For the data generating process considered here, this specification is correct, and $\varphi(\cdot)$ is the standard normal cumulative distribution function. As described in Section 3.2, the scaled parameter vector $(\tilde{\alpha}_1, \tilde{\delta}_0, \tilde{\delta}_1)$ is estimated by maximising (17). The maximum is obtained by the Nelder-Mead algorithm with multiple starting points. As starting points, we use the probit, logit, and linear probability model estimates, five random perturbation of each of these estimates, and twenty randomly drawn points. With $\hat{s}(X)$ and the estimate of $E[Z|X]$ in hand, the feasible estimator is calculated using the moment condition (11).

We consider sample sizes $n = 100, 500, 1000$ and 2000. For each sample size, we take 1000. For the design with both compliers and defiers, Table 5 presents summary statistics for the sampling

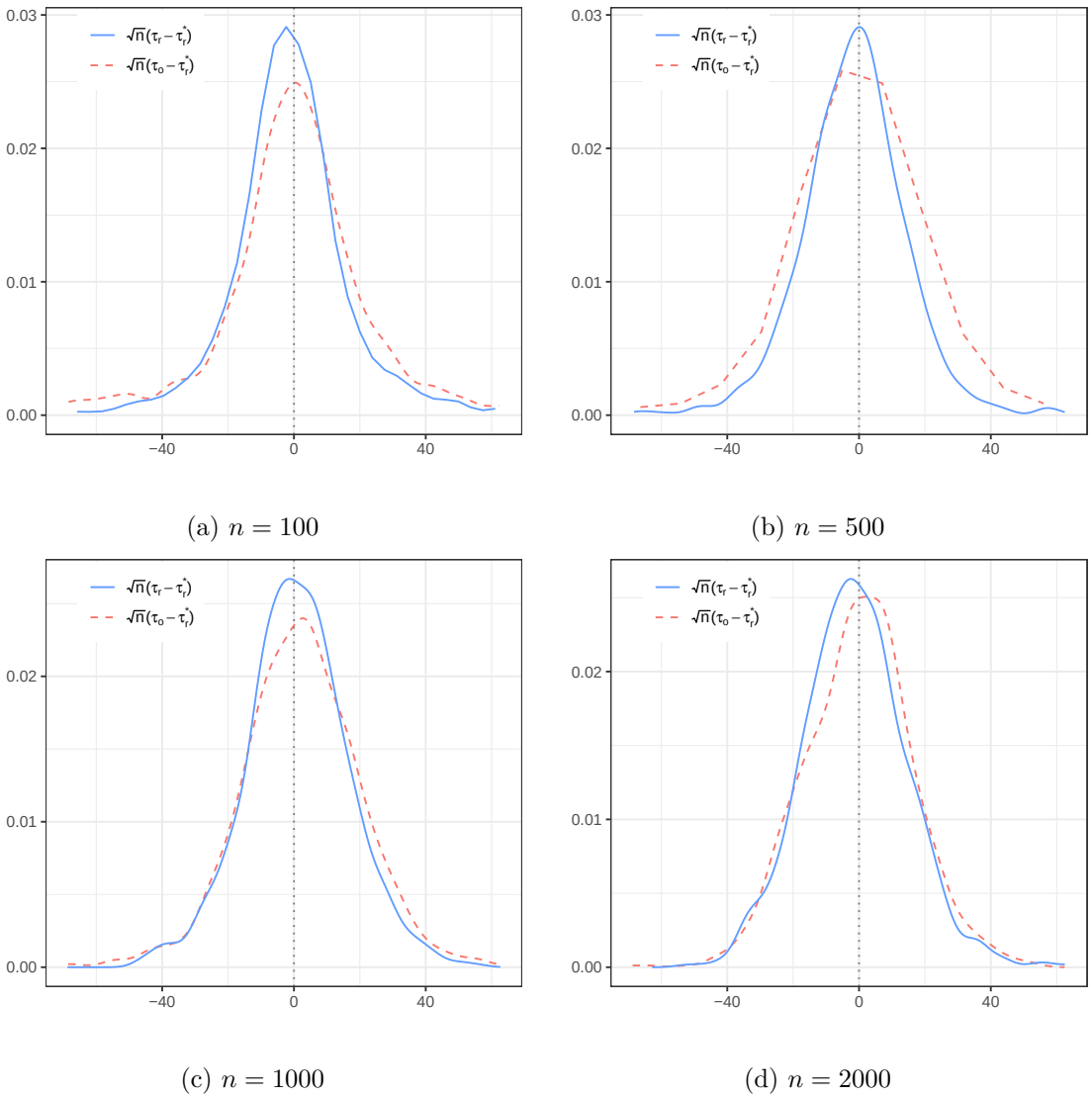
n_x	Mean	Standard Error	MSE	Median	First Quartile	Third Quartile
Oracle Estimator						
100	< -1,000,000	> 1,000,000	> 1,000,000	-0.038	-1.098	1.192
500	-0.422	8.795	77.452	-0.111	-0.620	0.423
1000	-0.036	0.578	0.338	-0.025	-0.381	0.331
2000	-0.100	0.375	0.141	-0.092	-0.346	0.126
Feasible Estimator						
100	-0.746	8.006	64.517	-0.267	-1.147	0.641
500	-0.193	0.876	0.775	-0.154	-0.590	0.240
1000	-0.070	0.500	0.251	-0.080	-0.372	0.240
2000	-0.127	0.353	0.126	-0.135	-0.353	0.086

Table 5: Design 1. Summary statistics for the sampling distribution of the oracle and feasible estimators. Results are for 1,000 draws.

	$\hat{\tau}_r$			$\hat{\tau}_o$		
n_x	90%	95%	99%	90%	95%	99%
100	0.937	0.968	0.974	0.729	0.786	0.832
500	0.931	0.971	0.998	0.854	0.912	0.954
1000	0.900	0.958	0.990	0.849	0.922	0.960
2000	0.890	0.947	0.987	0.886	0.947	0.970

Table 6: Design 1. Coverage rates of the 90%, 95% and 99% confidence intervals. Results are for 1,000 draws.

distribution, Table 6 presents coverage rates for the standard errors, and Figure 2 plots densities. Results are similar to those for the saturated specification of Section 2.4. The sampling distribution of the oracle estimator is more dispersed than that of the feasible estimator, but the two converge as the sample size increases. The MSE of the feasible estimator is lower than the oracle estimator, although the feasible estimator can be more biased. Results for the second design, where strong monotonicity holds, are presented in Tables 7 and 8 and Figure 3. These results are similar, although the performance of the feasible estimator is a little worse for the smaller sample sizes. However, as the sample size increases, there is little penalty for using the feasible estimator.



(a) $n = 100$ (b) $n = 500$

(c) $n = 1000$ (d) $n = 2000$

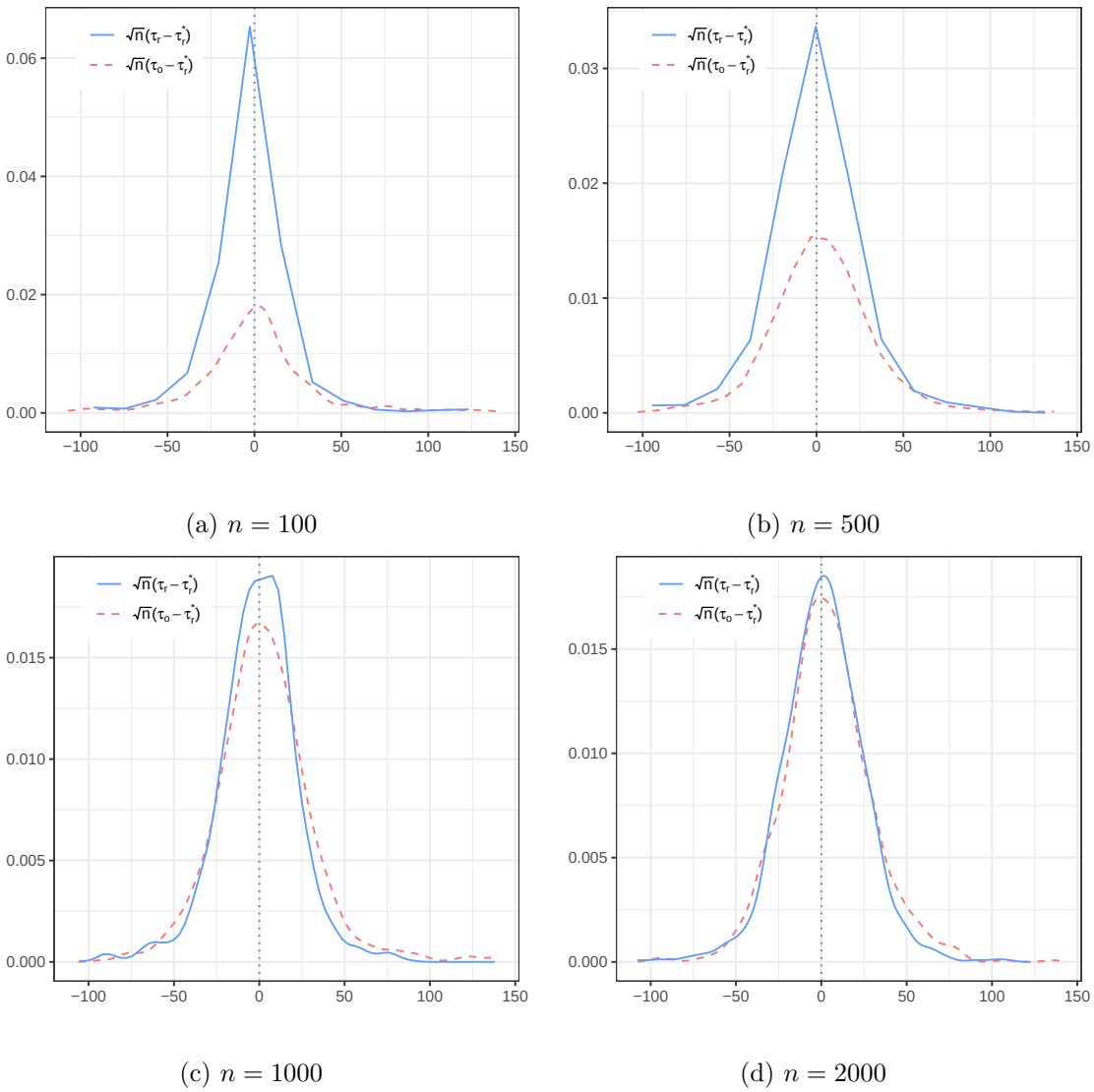
Figure 2: Design 1. Densities of $\sqrt{n}(\hat{\tau}_r - \tau_r^*)$ and $\sqrt{n}(\hat{\tau}_o - \tau_r^*)$. n is the sample size. Densities are calculated using results from 1,000 draws and the rule-of-thumb bandwidth.

n_x	Mean	Standard Error	MSE	Median	First Quartile	Third Quartile
Oracle Estimator						
100	$>$ 1,000,000	$>$ 1,000,000	$>$ 1,000,000	-0.060	-1.732	1.469
500	-0.011	6.502	42.287	0.006	-0.715	0.775
1000	0.013	1.283	1.660	-0.046	-0.537	0.467
2000	-0.020	0.577	0.339	-0.039	-0.371	0.303
Feasible Estimator						
100	-0.569	23.006	529.515	-0.311	-1.167	0.490
500	-0.391	10.890	118.681	-0.103	-0.663	0.432
1000	-0.085	1.758	3.091	-0.100	-0.530	0.298
2000	-0.075	0.533	0.285	-0.071	-0.395	0.258

Table 7: Design 2. Summary statistics for the sampling distribution of the oracle and feasible estimators. Results are for 1,000 draws.

	$\hat{\tau}_r$			$\hat{\tau}_o$		
n_x	90%	95%	99%	90%	95%	99%
100	0.917	0.927	0.929	0.617	0.659	0.699
500	0.970	0.981	0.983	0.801	0.851	0.882
1000	0.966	0.994	0.997	0.853	0.910	0.948
2000	0.938	0.980	0.998	0.879	0.936	0.972

Table 8: Design 2. Coverage rates of the 90%, 95% and 99% confidence intervals. Results are for 1,000 draws.



(a) $n = 100$

(b) $n = 500$

(c) $n = 1000$

(d) $n = 2000$

Figure 3: Design 2. Densities of $\sqrt{n}(\hat{\tau}_r - \tau_r^*)$ and $\sqrt{n}(\hat{\tau}_o - \tau_r^*)$. n is the sample size. Densities are calculated using results from 1,000 draws and the rule-of-thumb bandwidth.

5 Empirical Application

In this section, we apply the feasible reordered IV estimator to the design of Card (1993), which estimates the returns to education. The outcome Y is log hourly wage in 1976. D is years of education. We binarise D with the threshold set to 16 years of education. Z is a binary variable that equals one for individuals who lived close to an accredited four-year college in 1966.[8] Baseline controls are potential experience (defined as age − years of education − 6), a race dummy, and dummies for residence in the south and a standard metropolitan statistical area (SMSA) in 1976. A second specifications adds a dummy for residence in the South in 1966, and fixed effects for region of residence in 1966. The third further controls for each parents years of education. The fourth introduces a categorical variable constructed from both parents years of education. The final specification then adds controls for family structure at age 14.

We estimate each specification by two-stage least squares, and feasible reordered IV. The sample consists of 3,010 observations. When all conditioning covariates are included, there are 2,101 unique covariate combinations in the sample. As such, the saturate and control specification of Section 2.4 is unlikely to perform well, and we implement the semiparametric estimator proposed in Section 3. The expectation of Z is specified as a linear function of all covariates, For the propensity score, we use the specification of Assumption 6. The MRC estimates are obtained by the Nelder-Mead polytype algorithm with multiple starting points. As in the simulation exercise of Section 4, the starting points are the probit, logit and linear probability estimates, five random perturbations of each, and twenty random points. Standard errors are calculated using the formula (13).

Results are presented in Table 9. Each column corresponds to estimates for one set of controls. Standard errors are in paretheses. The two-stage least squares estimates are stable across specifications. For the second and fifth specifications, estimates are significant at the 10% level. In contrast, the reordered IV estimates are unstable across specifications, and none are significant. Regardless of the variation used, estimates for the first two specifications are imprecise.

6 Conclusion

This paper studies the reordered IV estimator suggested in Słoczyński (2026), which allows for both compliers and defiers to be present. Implementing this estimator requires estimating for each covariate value whether switchers are compliers or defiers. We showed that, if these estimates are $\sqrt{n}$-consistent and a margin condition holds, then this estimation is costless in terms of asymptotic variance. We proposed a general estimator where switcher type is estimated semiparametrically, and

[8] Specifically, if their local labour market in 1966 contained an accredited four year college.

Estimator					
TSLS	1.786	1.392*	1.460	1.574	1.544*
	(1.147)	(0.836)	(0.893)	(0.959)	(0.889)
Reordered IV	-1.257	0.492	-2.558	-3.155	-1.591
	(5.569)	(2.108)	(2.720)	(2.880)	(2.435)
Region Controls	No	Yes	Yes	Yes	Yes
Parental Education	No	No	Yes	Yes	Yes
Interacted Parental Education	No	No	No	Yes	Yes
Family Composition	No	No	No	No	Yes

Table 9: Estimates

showed that it is $\sqrt{n}$-consistent and asymptotically normal. In a simulation exercise, we found that the reordered IV estimator performs well, and resolves the negative weights problem encountered by conventional two-stage least squares. We applied the estimator to the data of Card (1993), and found that the difference between the conventional two-stage least squares and reordered IV estimates was large.

The methods presented in this paper could be further developed in two ways. First, they can be extended to multivalued and continuous treatments and instruments. Second, this paper has adopted the rich covariates assumption that the expectation of Z is a linear function of X. Modern estimators, such as that proposed in Chernozhukov et al. (2018) relax this assumption and use flexible machine learning methods to approximate the expectation function. Incorporating these flexible methods into the reordered IV estimator may be a promising direction for future research.

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A Omitted Derivations

A.1 Formula for τ_{tsls}

The denominator of (5) is decomposed as follows

$$\begin{aligned}
E[D(Z - L[Z|X = x])] &= E[D(Z - E[Z|X = x])] \\
&= E[\Pr(Z = 1|X = x)(1 - E[Z|X = x])E[D|X = x, Z = 1] \\
&\quad - \Pr(Z = 0|X = x)E[Z|X = x]E[D|X = x, Z = 0]] \\
&= E[\mathrm{Var}(Z|X = x)(E[D(1)|X = x, Z = 1] - E[D(0)|Z = 0, X = x])] \\
&= E[\mathrm{Var}(Z|X = x)(E[D(1)|X = x] - E[D(0)|X = x])] \\
&= E[\mathrm{Var}(Z|X = x)(E[D(1) - D(0)|X = x])] \\
&= E[s^*(X)\omega(x)]
\end{aligned}$$

The first equality uses Assumption 2. The second uses the law of iterated expectations. The third uses that Z is binary, so $E[Z|X] = \Pr(Z = 1|X)$ and $\Pr(Z = 1|X)\Pr(Z = 0|X) = \mathrm{Var}(Z|X)$. The fourth equality follows from Assumption 1 (ii). As $D(Z)$ and Z are independent conditional on X, $E[D(z)|Z = z, X = x] = E[D(z)|X = x]$. The final equality holds by the definitions of $\omega(x)$ and $s^*(X)$. Under Assumption 1 (iii) $E[D(1) - D(0)|X = x]$ is either $\Pr(D(1) > D(0)|X = x)$ or $-\Pr(D(0) > D(1)|X = x)$, which is equivalent to $\Pr(D(1) \neq D(0)|X = x)$ multiplied by $s^*(X)$.

For the numerator of (5)

$$\begin{aligned}
E[Y(Z - L[Z|X = x])] &= E[Y(Z - E[Z|X = x])] \\
&= E[(Y(0) + D(Y(1) - Y(0)))(Z - E[Z|X = x])] \\
&= E[E[D(Y(1) - Y(0))(Z - E[Z|X = x]|X = x]] \\
&= E[\mathrm{Var}(Z|X = x)(E[D(Y(1) - Y(0))|X = x, Z = 1] \\
&\quad - E[D(Y(1) - Y(0))|X = x, Z = 0])] \\
&= E[\mathrm{Var}(Z|X = x)E[(D(1) - D(0))(Y(1) - Y(0))|X = x]] \\
&= E[s^*(X)\omega(x)\tau(x)]
\end{aligned}$$

The first equality uses Assumption 2. The second follows as $Y = DY(1) + (1 - D)Y(0)$. The third holds as $E[Y(0)(Z - E[Z|X = x])|X = x] = E[Y(0)|X = x]E[Z - E[Z|X = x]|X = x] = 0$, where we use the conditional independence of $Y(0)$ and Z. For the fifth equality, we have $E[D(Y(1) - Y(0))|X = x, Z = z] = E[D(z)(Y(1) - Y(0))|X = x, Z = z]$, which conditional independence then implies is equal to $E[D(z)(Y(1) - Y(0))|X = x]$. By Assumption 1 (iii), $E[(D(1) - D(0))(Y(1) - Y(0))|X = x]$ is either $\Pr(D(1) > D(0)|X = x)E[Y(1) - Y(0)|X = x, D(1) > D(0)]$ or $-\Pr(D(0) >$

$D(1)|X = x)E[Y(1) - Y(0)|X = x, D(0) > D(1)]$, which is equivalent to $\Pr(D(1) \neq D(0)|X = x)\tau(x)$ multiplied by $s^*(X)$.

A.2 Derivation of τ_r

Consider the denominator of (8)

$$\begin{aligned}
E[Ds^*(X)(Z - L[Z|X = x])] &= E[Ds^*(X)(Z - E[Z|X = x])] \\
&= E[\Pr(Z = 1|X = x)s^*(X)(1 - E[Z|X = x])E[D|X = x, Z = 1] \\
&\quad - \Pr(Z = 0|X = x)s^*(X)E[Z|X = x]E[D|X = x, Z = 0]] \\
&= E[\mathrm{Var}(Z|X = x)s^*(X)(E[D(1)|X = x, Z = 1] - E[D(0)|Z = 0, X = x])] \\
&= E[\mathrm{Var}(Z|X = x)s^*(X)(E[D(1)|X = x] - E[D(0)|X = x])] \\
&= E[\mathrm{Var}(Z|X = x)s^*(X)(E[D(1) - D(0)|X = x])] \\
&= E[s^*(X)^2\omega(x)] \\
&= E[\omega(x)]
\end{aligned}$$

The penultimate equality uses that $E[D(1) - D(0)|X = x] = \Pr(D(1) > D(0)|X = x) - \Pr(D(1) < D(0)|X = x) = s^*(X)|p^*(x, 1) - p^*(x, 0)|$.

For the numerator of (8),

$$\begin{aligned}
E[Ys^*(X)(Z - L[Z|X = x])] &= E[Ys^*(X)(Z - E[Z|X = x])] \\
&= E[(Y(0) + D(Y(1) - Y(0)))s^*(X)(Z - E[Z|X = x])] \\
&= E[E[D(Y(1) - Y(0))(Z - E[Z|X = x]|X = x]] \\
&= E[\mathrm{Var}(Z|X = x)s^*(X)(E[D(Y(1) - Y(0))|X = x, Z = 1] \\
&\quad - E[D(Y(1) - Y(0))|X = x, Z = 0])] \\
&= E[\mathrm{Var}(Z|X = x)s^*(X)E[(D(1) - D(0))(Y(1) - Y(0))|X = x]] \\
&= E[s^*(X)^2\omega(x)\tau(x)] \\
&= E[\omega(x)\tau(x)]
\end{aligned}$$

B Proofs

B.1 Additional Assumptions

The following Assumptions collects regularity conditions needed for the proofs of Theorems 1, 2 and 3.

Assumption 8 (Theorem 1 Regularity Conditions)

(i) *The data* (X, Y, D, Z) *is an iid sample.*

(ii) $E[Ds^*(X)(Z - X^T\gamma^*)] > 0$

(iii) $E[XX^T] = \Sigma_X$*, where* Σ_X *is positive definite.*

(iv) $\gamma \in \Gamma$ *where* Γ *is a compact subset of* $\mathbb{R}^{d_x}$*.* $\tau_r \in \mathcal{T}_r$*, where* $\mathcal{T}_r$ *is a compact subset of* $\mathbb{R}$*.*

(v) *The moments* $E[||X||^2]$*,* $E[Y^2]$ *and* $E[Y^2 \cdot ||X||^2]$ *are finite.*

(vi) *There exists a universal constant* C *such that for any* $t \in [0, 1]$*, the expectations of* $||X||$*,* $|Y| \cdot ||X||$*,* Y^2 *and* $|\tau(X)|$ *conditional on* $|\Delta p^*(X)| \leq t$ *are bounded by* C*.*

Assumption 9 (Theorem 2 Regularity Conditions)

(i) $E[||X||^4] < \infty$

(ii) $E[|Y| \cdot ||X||^3] < \infty$

B.2 Notation

Let $E_n[\cdot]$ be the empirical expectation operator and $\mathbb{G}_n(\cdot)$ be the empirical process operator. For a generic variable X, $E_n[X] = \sum_{i=1}^n X_i/n$ and $\mathbb{G}_n(X) = \sqrt{n}(E_n(X) - E(X))$. Let $||\cdot||_{\mathcal{F}}$ denote the Frobenius norm, so that for a matrix A, $||A||_{\mathcal{F}} = \sqrt{\text{trace}(AA^T)}$. C is used to denote a genetic constant.

Define the errors

$$e_Y = Y - \tau_r^* D,$$
$$e_Z = Z - X^T\gamma^*,$$

and the moment conditions,

$$m^{e_Y}(s, \gamma) = e_Y s(X)(Z - X^T\gamma),$$
$$m^D(s, \gamma) = Ds(X)(Z - X^T\gamma).$$

Notice that

$$\tau_r^* = \frac{E[Ys^*(X)(Z - X^T\gamma^*)]}{E[Ds^*(X)(Z - X^T\gamma^*)]} \implies 0 = \frac{E[e_Y s^*(X)(Z - X^T\gamma^*)]}{E[Ds^*(X)(Z - X^T\gamma^*)]} \implies E[m^{e_Y}(s^*, \gamma^*)] = 0. \quad (22)$$

Finally, write

$$\Delta p(x) = p^*(x, 1) - p^*(x, 0).$$

B.3 Proof of Theorem 1

Under Assumption 2, $\hat{\gamma}$ is the linear projection of Z on X. Let $\mathbf{X}$ be a matrix formed by stacking X, $\Sigma_X = \mathbf{X}^T\mathbf{X}$, and $\mathbf{e}_Z$ is a vector that collects e_Z. Then $(\hat{\gamma} - \gamma^*) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{e}_Z$. From Assumption 8 (i) and (iv),

$$\sqrt{n}(\hat{\gamma} - \gamma^*) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{e}_Z/\sqrt{n} + o_p(1), \tag{23}$$

where $\mathbf{X}^T\mathbf{e}_Z/\sqrt{n} = O_p(1)$ by the Lindeberg-Feller CLT.

Now consider the oracle estimator. $\sqrt{n}(\hat{\tau}_o - \tau_r^*)$ can be decomposed as

$$\begin{aligned}\sqrt{n}(\hat{\tau}_o - \tau_r^*) &= \sqrt{n}\frac{E_n[m^{e_Y}(s^*, \hat{\gamma})]}{E_n[m^D(s^*, \hat{\gamma})]}, \\ &= \frac{\mathbb{G}_n[m^{e_Y}(s^*, \hat{\gamma})] + \sqrt{n}E[m^{e_Y}(s^*, \hat{\gamma}) - m^{e_Y}(s^*, \gamma^*)]}{E_n[m^D(s^*, \hat{\gamma})]},\end{aligned}$$

where the second line uses (22). Lemma 1 establishes the convergence of the empirical process $\mathbb{G}_n(m^{e_Y}(s^*, \hat{\gamma})$ to $\mathbb{G}_n(m^{e_Y}(s^*, \hat{\gamma}^*)$. Using this, the form of $m^{e_Y}(s, \gamma)$, and (23),

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) = \frac{\mathbb{G}_n(m^{e_Y}(s^*, \gamma^*)) - E[e^Y s^*(X)X]^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{e}_Z/\sqrt{n}}{E_n[m^D(s^*, \hat{\gamma})]} + o_p(1) \tag{24}$$

Furthermore, using (22) and the definition of e_Z

$$\mathbb{G}_n(m^{e_Y}(s^*, \gamma^*)) = \sqrt{n}\left(E_n[m^{e_Y}(s^*, \gamma^*)] - E[m^{e_Y}(s^*, \gamma^*)]\right) = \sqrt{n}E_n[e_Y s^*(X)e_Z],$$

so

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) = \frac{\sqrt{n}E_n[e_Y s^*(X)e_Z] - E[e_Y s^*(X)X]^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{e}_Z/\sqrt{n}}{E_n[m^D(s^*, \hat{\gamma})]} + o_p(1). \tag{25}$$

The vector $(e_Y s^*(X)e_Z, Xe_Z)$ is mean zero and has covariance matrix

$$\Omega = E\left[\begin{pmatrix} e_Y s^*(X)e_Z \\ Xe_Z \end{pmatrix}\begin{pmatrix} e_Y s^*(X)e_Z \\ Xe_Z \end{pmatrix}^T\right]$$

By the Cramer-Wold device and the Lindeberg-Feller CLT, $(e_Y s^*(X)e_Z, Xe_Z)$ is normally distributed. By Lemma 2, $E_n[m^D(s^*, \hat{\gamma})] \xrightarrow{p} E[m^D(s^*, \gamma^*)]$. Applying Slutsky's Theorem to (25),

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) \xrightarrow{d} N(0, \sigma_{\tau_r})$$

with

$$\sigma_{\tau_r}^2 = E[m^D(s^*, \gamma^*)]^{-2}\begin{pmatrix} 1 \\ -E[e_Y s^*(X)X]^T\Sigma_X^{-1} \end{pmatrix}^T \Omega \begin{pmatrix} 1 \\ -E[e_Y s^*(X)X]^T\Sigma_X^{-1} \end{pmatrix}$$

For the feasible estimator, $\sqrt{n}(\hat{\tau}_r - \tau_r^*)$, can be decomposed as

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) = \frac{\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) + \sqrt{n}E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)]}{E_n[m^D(\hat{s},\hat{\gamma})]}$$

By Lemma 3, $\sqrt{n}(E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)]) = \sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma}-\gamma^*) + o_p(1)$ and $\sqrt{n}(E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)]) = \sqrt{n}E[e_Y s^*(X)X]^T(\hat{\gamma}-\gamma^*) + o_p(1)$. Lemma 5 shows that $\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) = \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) + o_p(1)$. Hence

$$\sqrt{n}(\hat{\tau}_r - \tau_r^*) = \frac{\left(\mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) - E[e_Y s^*(X)X]^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{e}_Z/\sqrt{n}\right)}{E_n[m^D(\hat{s},\hat{\gamma})]}$$

which differs from the oracle estimator only in that $E[m^D(\hat{s},\hat{\gamma})]$ replaces $E[m^D(s^*,\hat{\gamma})]$. Lemmas 3 and 5 establish the conditions needed for Lemma 6. The result follows by applying this lemma and the same argument as the oracle estimator.

Lemma 1 *Suppose that Assumptions 1, 2, and 8 hold*

(i) $\mathbb{G}_n(m^{e_Y}(s^*,\hat{\gamma}) = \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) + o_p(1)$

(ii) $\mathbb{G}_n(m^D(s^*,\hat{\gamma}) = \mathbb{G}_n(m^D(s^*,\gamma^*)) + o_p(1)$

For claim (i), first write

$$\begin{aligned}
&\mathbb{G}_n(m^{e_Y}(s^*,\hat{\gamma})) - \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) \\
&= \sqrt{n}\left(E_n[m^{e_Y}(s^*,\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] - E[m^{e_Y}(s^*,\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)]\right) \\
&= \sqrt{n}(E_n[e_Y s^*(X)X] - E[e_Y s^*(X)X])^T(\gamma^* - \hat{\gamma})
\end{aligned}$$

By Assumption (8),

$$\begin{aligned}
&||E[e_Y s^*(X)X]|| \leq E[||e_Y||X] \leq E[|Y| \cdot ||X||] + \tau_r^* E[||X||] < \infty \\
&||E[e_Y s^*(X)X(e_Y s^*(X)X)^T]||_{\mathcal{F}} \leq E[e_Y^2||XX^T||_{\mathcal{F}}] \leq E[Y^2||X||^2] + \tau_r^{*2}E[||X||^2] < \infty
\end{aligned}$$

Hence, $E_n[e_Y s^*(X)X] - E[e_Y s^*(X)X]$ is an average of mean zero vectors with a finite covariance matrix. By the Lindeberg-Levy CLT, $\sqrt{n}(E_n[Ds^*(X)X] - E[Ds^*(X)X])$ is $O_p(1)$. The claim follows as $||\hat{\gamma} - \gamma^*|| = o_p(1)$.

The argument for claim (ii) follows the same steps. In this case, we apply a CLT to $\sqrt{n}(E_n[Ds^*(X)X] - E[Ds^*(X)X])$. Since $|Ds^*(X)X| \leq |X|$, the finiteness of the mean and covariance matrix is immediate.

Lemma 2 *Suppose that Assumptions 1, 2, and 8 hold*

$$E_n[m^D(s^*,\hat{\gamma})] = E[m^D(s^*,\gamma^*)] + o_p(1)$$

$$
\begin{aligned}
E_n[m^D(s^*,\hat{\gamma})] - E[m^D(s^*,\gamma^*)] &= E_n[m^D(s^*,\hat{\gamma}) - E[m^D(s^*,\hat{\gamma})]] + E[m^D(s^*,\hat{\gamma}) - m^D(s^*,\gamma^*)] \\
&= n^{-1/2}\mathbb{G}_n(m^D(s^*,\hat{\gamma})) + n^{-1/2}\sqrt{n}E[m^D(s^*,\hat{\gamma}) - m^D(s^*,\gamma^*)] \\
&= n^{-1/2}\mathbb{G}_n(m^D(s^*,\gamma^*)) + n^{-1/2}E[Ds^*(X)X]^T\sqrt{n}(\gamma^* - \hat{\gamma}) + o_p(1) \\
&= n^{-1/2}O_p(1) + n^{-1/2}O_p(1) + o_p(1) \\
&= o_p(1)
\end{aligned}
$$

The third line uses the result of Lemma 1. For the fourth, $\mathbb{G}_n(m^D(s^*,\gamma^*))$ is $O_p(1)$ by the Lindeberg-Levy CLT, $\sqrt{n}(\hat{\gamma} - \gamma^*)$ is $O_p(1)$ by the Lindeberg-Feller CLT, and $||E[Ds^*(X)X]|| \leq E[||X||] < \infty$ by Assumption 8.

Lemma 3 *Suppose that Assumptions 1 - 5 and 8 hold*

(i) $\sqrt{n}E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] = -\sqrt{n}E[e_Y s^*(X)X]^T(\hat{\gamma} - \gamma^*) + o_p(1)$

(ii) $\sqrt{n}E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)]) = -\sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma} - \gamma^*) + o_p(1)$

For claim (i),

$$
\sqrt{n}E[m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] = \sqrt{n}E[m^{e_Y}(s^*,\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*)] - \sqrt{n}\mathcal{A}_1 + \sqrt{n}\mathcal{A}_2
$$

with

$$
\begin{aligned}
\mathcal{A}_1 &= E[e_Y(\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)], \\
\mathcal{A}_2 &= E[e_Y(\hat{s}(X) - s^*(X))(Z - X^T\gamma^*)],
\end{aligned}
$$

Each of these terms is $o_p(1)$. For $\mathcal{A}_1$,

$$
\begin{aligned}
|\mathcal{A}_1| &\leq E[|e_Y(\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)|] \\
&\leq E[(|Y| + |\tau_r^*|D)\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}|X^T(\hat{\gamma} - \gamma^*)|] \\
&\leq E[\mathbb{1}\{|\Delta p^*(X)| \leq 2||\hat{p} - p^*||_\infty\}(|Y| + |\tau_r^*|)||X||] \cdot ||\hat{\gamma} - \gamma^*|| \\
&\leq C\Pr(|\Delta p^*(x)| \leq 2||\hat{p} - p^*||_\infty)||\hat{\gamma} - \gamma^*|| \\
&\leq C||\hat{p} - p^*||_\infty^\zeta||\hat{\gamma} - \gamma^*|| \\
&= O_p(n^{-(1+\zeta)/2}).
\end{aligned}
$$

The third line holds as $X^T(\hat{\gamma} - \gamma^*)$ is a scalar, so $|X^T(\hat{\gamma} - \gamma^*)| = ||X^T(\hat{\gamma} - \gamma^*)||$. The Cauchy-Schwarz inequality then gives the inequality. The fourth line uses Assumption 8. As $\zeta > 0$, $\sqrt{n}|\mathcal{A}_1| = O_p(n^{-\zeta/2}) = o_p(1)$.

For $\mathcal{A}_2$,

$$\begin{aligned}
|\mathcal{A}_2| &= |E[(\hat{s}(X) - s^*(X))E[e_Y(Z - X^T\gamma^*)|X]]| \\
&= |E[(\hat{s}(x) - s^*(x))\text{Var}(Z|X = x)\Delta p^*(x)(\tau(x) - \tau_r^*)]| \\
&\leq CE[\mathbb{1}\{\hat{s}(x) \neq s^*(x)\}|\Delta p^*(x)|(|\tau(x)| + |\tau_r^*|)] \\
&\leq CE[\mathbb{1}\{|\Delta p^*(x)| \leq 2||\hat{p} - p^*||_\infty\}|\Delta p^*(x)|(|\tau(x)| + |\tau_r^*|)] \\
&\leq C||\hat{p} - p^*||_\infty \Pr(|\Delta p^*(x)| \leq 2||\hat{p} - p^*||_\infty) \\
&\leq C||\hat{p} - p^*||_\infty^{1+\zeta} \\
&= O_p(n^{-(1+\zeta)/2}).
\end{aligned}$$

The second line follows from the first as

$$E[e_Y(Z - X^T\gamma^*)|X] = E[Y(Z - X^T\gamma^*)|X] - \tau_r^* E[D(Z - X^T\gamma^*)|X] = \text{Var}(Z|X)\Delta p^*(X)(\tau(X) - \tau_r^*).$$

The fifth line uses $\tau(x)$ Assumption 8. It follows that $\sqrt{n}\mathcal{A}_2 = O_p(n^{-\zeta/2}) = o_p(1)$.

For claim (ii), the decomposition is similar. The additional terms are

$$\begin{aligned}
\mathcal{B}_1 &= E[D(\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)] \\
\mathcal{B}_2 &= E[D(\hat{s}(X) - s^*(X))(Z - X^T\gamma^*)]
\end{aligned}$$

For $\mathcal{B}_1$,

$$|\mathcal{B}_1| = |E[D(\hat{s}(X) - s^*(X))X^T(\hat{\gamma} - \gamma^*)]| \leq E[|\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}|X^T(\hat{\gamma} - \gamma^*)|].$$

The remaining steps are identical to the proof for $\mathcal{A}_1$.

For $\mathcal{B}_2$,

$$|\mathcal{B}_2| = |E[(\hat{s}(X) - s^*(X))E[D(Z - X^T\gamma^*)|X]]| \leq E[\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}|\Delta p^*|]$$

The claim follows by the same argument as $\mathcal{A}_2$.

Lemma 4 *Suppose that Assumptions 2, 4, and 8 hold.*

(i) Consider the pseudometric

$$\rho^{e_Y}((s,\gamma),(s',\gamma')) = E[(m^{e_Y}(s,\gamma) - m^{e_Y}(s',\gamma'))^2]^{1/2}$$

For every $\eta > 0, \epsilon > 0$, there exists $\xi > 0$ such that

$$\lim_{n\to\infty} \Pr\left(\sup_{(s,s')\in\mathcal{S}^2,(\gamma,\gamma')\in\Gamma^2:\rho^{e_Y}((s,\gamma),(s',\gamma'))<\xi} |\mathbb{G}_n(m^{e_Y}(s,\gamma) - \mathbb{G}_n(m^{e_Y}(s',\gamma'))| > \eta\right) < \epsilon$$

(ii) Consider the pseudometric

$$\rho^D((s,\gamma),(s',\gamma')) = E[(m^D(s,\gamma) - m^D(s',\gamma'))^2]^{1/2}$$

For every $\eta > 0, \epsilon > 0$, there exists $\xi > 0$ such that

$$\lim_{n\to\infty} \Pr\left(\sup_{(s,s')\in\mathcal{S}^2,(\gamma,\gamma')\in\Gamma^2:\rho^D((s,\gamma),(s',\gamma'))<\xi} |\mathbb{G}_n(m^D(s,\gamma) - \mathbb{G}_n(m^D(s',\gamma'))| > \eta \right) < \epsilon$$

Both claims can be shown by verifying Assumptions A, B, and C of Theorem 1 of Andrews (1994). Assumption C is satisfied as the data are i.i.d. To verify Assumptions A and B for claim (i), consider the function class

$$\mathcal{M}^{e_Y} \coloneqq \{m^{e_Y}(s,\gamma) : s \in \mathcal{S}, \gamma \in \Gamma\},$$

Notice that

$$|(Y - \tau_r^* D)s(X)(Z - X^T\gamma)| = |Y - \tau_r^* D||Z - X^T\gamma| \leq (|Y| + C)(1 + ||X||C)$$

where we use that Γ is compact and τ_r^* is finite. Hence, this function class possesses a measurable envelope. Assumption B then holds as the data are i.i.d. For Assumption A, any $m^{e_Y}(s,\gamma)$ can be written as $f \times s$ for some f in the function class $\mathcal{F} \coloneqq \{(Y - \tau_r^* D)(Z - X^T\gamma) : \gamma \in \Gamma\}$ and $s \in \mathcal{S}$, with $\mathcal{S}$ as defined in Assumption 4. $\mathcal{F}$ is a finite-dimensional vector space of measurable functions so, by Lemma 2.6.16 of Vaart and Wellner (2023), it is a VC-subgraph class of functions. $\mathcal{S}$ is a VC-class by Assumption 4. By Theorem 2.6.7 of Vaart and Wellner (2023), the uniform covering numbers of $\mathcal{S}$ and $\mathcal{F}$ are finite, so they satisfy Assumption A. Theorem 3 of Andrews (1994) then implies that $m^D(s,\gamma)$ satisfies Assumption A.

For claim (ii), consider the function class

$$\mathcal{M}^D = \{m^D(s,\gamma) : s = \mathbb{1}\{p^*(x,1) > p^*(x,0)\}, p \in \mathcal{P}, \gamma \in \Gamma\}.$$

The envelope for this function class is $(1 + ||X||C)$, and Assumptions A and B can be verified in the same manner.

Lemma 5 *Suppose that Assumptions 1 - 5 and Assumption 8 hold.*

(i) $\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) = \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)) + o_p(1)$

(ii) $\mathbb{G}_n(m^D(\hat{s},\hat{\gamma})) = \mathbb{G}_n(m^D(s^*,\gamma^*)) + o_p(1)$

For claim (i), consider $E[(m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*))^2]$.

$$\begin{aligned}
E[(m^{e_Y}(\hat{s},\hat{\gamma}) - m^{e_Y}(s^*,\gamma^*))^2] &= E[e_Y^2(\hat{s}(X)X^T(\gamma^* - \hat{\gamma}) + (\hat{s}(X) - s^*(X))(Z - X^T\gamma^*))^2] \\
&\leq CE[e_Y^2(X^T(\gamma^* - \hat{\gamma}))^2] + CE[e_Y^2((\hat{s}(X) - s^*(X))(Z - X^T\gamma^*))^2] \\
&\leq CE\left[(Y^2 + \tau_r^{*2})||X||^2\right]||\gamma - \hat{\gamma}||^2 + CE\left[(Y^2 + \tau_r^{*2})\mathbb{1}\{\hat{s}(X) \neq s(X)\}\right] \\
&\leq C||\gamma - \hat{\gamma}||^2 + C\Pr\left(|\Delta p^*(x)| \leq 2||\hat{p} - p^*||_\infty\right) \\
&\leq C||\gamma - \hat{\gamma}||^2 + C||\hat{p} - p^*||_\infty^\zeta \\
&= o_p(1).
\end{aligned}$$

The third inequality holds as $Z - X^T\gamma^* = Z - E[Z|X]$, which is bounded in magnitude by 1. The fourth line uses Assumption 8. For the fifth inequality, we use the magin condition. The last line holds as $||\hat{\gamma} - \gamma^*|| = o_p(1)$, $||\hat{p} - p||_\infty = o_p(1)$ and $\zeta > 0$. This implies that $\rho^D((\hat{s},\hat{\gamma}),(s^*,\gamma^*)) = o_p(1)$, where $\rho^{e_Y}(\cdot,\cdot)$ is a pseudometric introduced in Lemma 4. The remaining steps are a standard argument. Fix $\eta > 0$ and $\epsilon > 0$,

$$\begin{aligned}
&\Pr(|\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) - \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*))| > \eta) \\
&\leq \Pr(\rho^{e_Y}((\hat{s},\hat{\gamma}),(s^*,\gamma^*)) \geq \xi) + \Pr(|\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) - \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*)| > \eta, \rho^{e_Y}((\hat{s},\hat{\gamma}),(s^*,\gamma^*)) < \xi) \\
&\leq \Pr(\rho^{e_Y}((\hat{s},\hat{\gamma}),(s^*,\gamma^*)) \geq \xi) \\
&\quad + \Pr\left(\sup_{(s,s')\in\mathcal{S}^2,(\gamma,\gamma')\in\Gamma^2:\rho^{e_Y}((s,\gamma),(s',\gamma'))<\xi} |\mathbb{G}_n(m^{e_Y}(s,\gamma) - \mathbb{G}_n(m^{e_Y}(s',\gamma'))| > \eta\right).
\end{aligned}$$

The first inequality holds by the union bound. By the above argument and Lemma 4, there exist ξ and N such that, for all $n > N$, both terms are smaller than $\epsilon/2$. As this argument holds for any η and ϵ, $\mathbb{G}_n(m^{e_Y}(\hat{s},\hat{\gamma})) - \mathbb{G}_n(m^{e_Y}(s^*,\gamma^*))$ is $o_p(1)$.

For claim (ii), as D is binary,

$$\begin{aligned}
E[(m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*))^2] &= E[D^2(\hat{s}(X)X^T(\gamma^* - \hat{\gamma}) + (\hat{s}(X) - s^*(X))(Z - X^T\gamma^*))^2] \\
&\leq E[(\hat{s}(X)X^T(\gamma^* - \hat{\gamma}) + (\hat{s}(X) - s^*(X))(Z - X^T\gamma^*))^2],
\end{aligned}$$

and the rest of the argument proceeds as above.

Lemma 6 *Suppose that Assumption 8 holds. If*

$$\mathbb{G}_n(m^D(\hat{s},\hat{\gamma})) = \mathbb{G}_n(m^D(s^*,\gamma^*)) + o_p(1),$$

and

$$\sqrt{n}E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)]) = -\sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma} - \gamma^*) + o_p(1),$$

then

$$E_n[m^D(\hat{s},\hat{\gamma})] = E[m^D(s^*,\gamma^*)] + o_p(1).$$

$$
\begin{aligned}
E_n[m^D(\hat{s},\hat{\gamma})] - E[m^D(s^*,\gamma^*)] &= E_n[m^D(\hat{s},\hat{\gamma}) - E[m^D(\hat{s},\hat{\gamma})]] + E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)] \\
&= n^{-1/2}\mathbb{G}_n(m^D(\hat{s},\hat{\gamma})) + n^{-1/2}\sqrt{n}E[m^D(\hat{s},\hat{\gamma}) - m^D(s^*,\gamma^*)] \\
&= n^{-1/2}\mathbb{G}_n(m^D(s^*,\gamma^*)) - n^{-1/2}\sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma}-\gamma^*) + o_p(1) \\
&= n^{-1/2}O_p(1) + n^{-1/2}O_p(1) + o_p(1) \\
&= o_p(1)
\end{aligned}
$$

The third line uses the conditions stated in the Lemma. For the fourth, $\mathbb{G}_n(m^D(s^*,\gamma^*))$ is $O_p(1)$ by the Lindeberg-Levy CLT, and $\sqrt{n}(\hat{\gamma}-\gamma^*)$ is $O_p(1)$ by the Lindeberg-Feller CLT.

B.4 Proof of Theorem 2

Consider the numerator of (13). Expanding the quadratic gives three terms

$$
\begin{aligned}
\mathcal{C}_1 &= E_n[(\hat{e}_Y\hat{s}(X)\hat{e}_Z)^2] \\
\mathcal{C}_2 &= E_n[\hat{e}_Y\hat{s}(X)X]^T E_n[XX^T]^{-1}E_n[\hat{e}_Y\hat{s}(X)X\hat{e}_Z^2] \\
\mathcal{C}_3 &= E_n[\hat{e}_Y\hat{s}(X)X]^T E_n[XX^T]^{-1}E_n[XX^T\hat{e}_Z^2]E_n[XX^T]^{-1}E_n[\hat{e}_Y\hat{s}(X)X]
\end{aligned}
$$

By the law of large numbers, $E_n[XX^T]$ converges in probability to Σ_X. Lemmas 8 - 11 establish that $E_n[(\hat{e}_Y\hat{s}(X)\hat{e}_Z)^2]$, $E_n[\hat{e}_Y\hat{s}(X)X]$, $E_n[\hat{e}_Y\hat{s}(X)X\hat{e}_Z^2]$, and $E_n[XX^T\hat{e}_Z^2]$ are consistent estimators of $E[(e_Ys^*(X)e_Z)^2]$, $E[e_Ys^*(X)X]$, $E[e_Ys^*(X)Xe_Z^2]$, and $E[XX^Te_Z^2]$. Consistency of the numerator as a whole follows by the continuous mapping theorem. For the denominator of (13), Lemma 6 establishes the consistency of $\sum_{i=1}^n D_i\hat{s}(X_i)\hat{e}_{Zi}/n$, and the continuous mapping theorem then gives consistency of $(\sum_{i=1}^n D_i\hat{s}(X_i)\hat{e}_{Zi}/n)^{-2}$.

Lemma 7 *Suppose that Assumption 8 holds. The following are P-Glivenko-Cantelli classes of functions,*

$$
\begin{aligned}
\mathcal{F} &\coloneqq \{y - \tau_r d : \tau_r \in \mathcal{T}_r\} \\
\mathcal{G} &\coloneqq \{z - x^T\gamma : \gamma \in \Gamma\}.
\end{aligned}
$$

We first establish envelope functions for both function classes. As $\mathcal{T}_r$ is bounded, $|Y - \tau_r D| \leq |Y| + C$, and $F = |Y| + C$ is an envelope for $\mathcal{F}$. For $\mathcal{G}$, as Γ is bounded, $|Z - X^T\gamma| \leq 1 + ||X||C$, and $G = 1 + ||X||C$ is an envelope. By Assumption 8, $E[|Y|]$ and $E[||X||]$ are finite.

$\mathcal{F}$ and $\mathcal{G}$ are both finite dimensional vector spaces, and thus VC-subgraph class by Lemma 2.6.16 of Vaart and Wellner (2023). The result follows by Theorem 2.8.1 of Vaart and Wellner (2023).

Lemma 8 *Suppose that Assumptions 1 - 5 and Assumptions 8 - 9 hold.*

$$E_n[(\hat{e}_Y \hat{s}(X)\hat{e}_Z)^2] = E[(e_Y s^*(X)e_Z)^2] + o_p(1)$$

Trivially,

$$E_n[(\hat{e}_Y \hat{s}(X)\hat{e}_Z)^2] - E[(e_Y s^*(X)e_Z)^2] = E_n[(\hat{e}_Y \hat{e}_Z)^2] - E[(\hat{e}_Y \hat{e}_Z)^2] + E[(\hat{e}_Y \hat{e}_Z)^2] - E[(e_Y e_Z)^2].$$

$E_n[(\hat{e}_Y \hat{e}_Z)^2] - E[(\hat{e}_Y \hat{e}_Z)^2]$ is $o_p(1)$ by Lemma 7 and Theorem 2.10.5 of Vaart and Wellner (2023). The other term can be decomposed as,

$$E[(\hat{e}_Y \hat{e}_Z)^2] - E[(e_Y e_Z)^2] = E[\hat{e}_Y^2(\hat{e}_Z^2 - e_Z^2)] + E[(\hat{e}_Y^2 - e_Y^2)e_Z^2],$$

with

$$\begin{aligned}
|E[\hat{e}_Y^2(\hat{e}_Z^2 - e_Z^2)]| &= |2E[\hat{e}_Y^2 e_Z(\hat{e}_Z - e_Z)] + E[\hat{e}_Y^2(\hat{e}_Z - e_Z)^2]| \\
&\leq 2E[\hat{e}_Y^2|\hat{e}_Z - e_Z|] + E[\hat{e}_Y^2(\hat{e}_Z - e_Z)^2] \\
&\leq 2E[\hat{e}_Y^2||X||] \cdot ||\hat{\gamma} - \gamma^*|| + E[\hat{e}_Y^2||X||^2] \cdot ||\hat{\gamma} - \gamma^*||^2 \\
&\leq 2(2E[Y^2||X||] + CE[||X||]) \cdot ||\hat{\gamma} - \gamma^*|| + (2E[Y^2||X||^2] + CE[||X||]^2) \cdot ||\hat{\gamma} - \gamma^*||^2 \\
&= C||\hat{\gamma} - \gamma^*|| + C||\hat{\gamma} - \gamma^*||^2 \\
&= o_p(1) + o_p(1)
\end{aligned}$$

The second line holds by the triangle inequality, and as e_Z is bounded in magnitude by 1. The third line uses $\hat{e}_z - e_z = X^T(\gamma^* - \hat{\gamma})$. For the fourth line, $\hat{e}_Y^2 \leq 2(Y^2 + \hat{\tau}_r^2 D^2)$, which is bounded in magnitude by $2(Y^2 + C)$. The fifth line uses Assumption 9.

Finally,

$$\begin{aligned}
|E[(\hat{e}_Y^2 - e_Y^2)e_Z^2]| &\leq E[|\hat{e}_Y^2 - e_Y^2|] \\
&\leq 2E[|YD|]|\tau_r^* - \hat{\tau}_r| + E[D^2]|\hat{\tau}_r - \tau_r^*|^2 \\
&\leq 2E[|Y|]|\tau_r^* - \hat{\tau}_r)| + |\hat{\tau}_r - \tau_r^*|^2 \\
&= C|\tau_r^* - \hat{\tau}_r)| + |\hat{\tau}_r - \tau_r^*|^2 \\
&= o_p(1) + o_p(1)
\end{aligned}$$

The first line holds by the triangle inequality and as $|e_Z| \leq 1$. The third holds by Assumption 8 and as D is binary.

Lemma 9 *Suppose that Assumptions 1 - 5 and Assumptions 8 - 9 hold.*

$$E_n[\hat{e}_Y \hat{s}(X)X] = E[e_Y s^*(X)X] + o_p(1)$$

Write

$$E_n[\hat{e}_Y \hat{s}(X)X] - E[e_Y s^*(X)X] = E_n[\hat{e}_Y \hat{s}(X)X] - E[\hat{e}_Y \hat{s}(X)X] + E[(\hat{e}_Y \hat{s}(X) - e_Y s^*(X))X].$$

The first term is $o_p(1)$ by Lemma 7, Assumption 4 and Theorem 2.10.5 of Vaart and Wellner (2023). For the second,

$$\begin{aligned}
||E[(\hat{e}_Y \hat{s}(X) - e_Y s^*(X))X]|| &\leq E[||\hat{s}(X)(\hat{e}_Y - e_Y)X||] + E[||e_Y(\hat{s}(X) - s^*(X))X||] \\
&\leq ||\tau_r^* - \hat{\tau}_r|| E[||X||] + E[\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}||e_Y X||] \\
&\leq C||\tau_r^* - \hat{\tau}_r|| + CE[\mathbb{1}\{|\Delta p^*| \leq 2||\hat{p} - p^*||_\infty\}(||YX|| + C||X||)] \\
&\leq C||\tau_r^* - \hat{\tau}_r|| + C||\hat{p} - p^*||_\infty^\zeta \\
&= o_p(1) + o_p(1).
\end{aligned}$$

To obtain the fourth inequality, invoke Assumption 5 and Assumption 9.

Lemma 10 *Suppose that Assumptions 1 - 5 and Assumptions 8 - 9 hold.*

$$E_n[\hat{e}_Y \hat{s}(X)X\hat{e}_Z^2] = E[e_Y s^*(X)Xe_Z^2] + o_p(1)$$

Write

$$\begin{aligned}
E_n[\hat{e}_Y \hat{s}(X)X\hat{e}_Z^2] - E[e_Y s^*(X)Xe_Z^2] = E_n[\hat{e}_Y \hat{s}(X)X\hat{e}_Z^2] - E[\hat{e}_Y \hat{s}(X)X\hat{e}_Z^2] \\
+ E[\hat{e}_Y \hat{s}(X)X\hat{e}_Z^2] - E[e_Y s^*(X)Xe_Z^2].
\end{aligned}$$

By an identical argument to Lemma 9, the first term is $o_p(1)$. Turning to the second,

$$||E[\hat{e}_Y \hat{s}(X)X\hat{e}_Z^2] - E[e_Y s^*(X)Xe_Z^2]|| \leq ||E[\hat{e}_Y \hat{s}(X)X(\hat{e}_Z^2 - e_Z^2)]|| + ||E[(\hat{e}_Y \hat{s}(X) - e_Y s^*(X))Xe_Z^2]||.$$

Here

$$\begin{aligned}
||E[\hat{e}_Y \hat{s}(X)X(\hat{e}_Z^2 - e_Z^2)]|| &\leq E[||\hat{e}_Y X(2(\hat{e}_Z - e_Z) + (\hat{e}_Z - e_Z)^2)||] \\
&\leq 2E[||\hat{e}_Y|| \cdot ||X||^2] \cdot ||\gamma^* - \hat{\gamma}|| + E[||\hat{e}_Y|| \cdot ||X||^3] \cdot ||\gamma^* - \hat{\gamma}||^2 \\
&\leq 2E[(|Y| + C) \cdot ||X||^2] \cdot ||\gamma^* - \hat{\gamma}|| + E[(|Y| + C) \cdot ||X||^3] \cdot ||\gamma^* - \hat{\gamma}||^2 \\
&\leq C \cdot ||\gamma^* - \hat{\gamma}|| + C \cdot ||\gamma^* - \hat{\gamma}||^2 \\
&= o_p(1) + o_p(1),
\end{aligned}$$

where the fourth inequality holds by Assumption 9. Finally,

$$\begin{aligned}
||E[(\hat{e}_Y\hat{s}(X) - e_Y s^*(X))Xe_Z^2]|| &\leq ||E[\hat{s}(X)(\hat{e}_Y - e_Y)Xe_Z^2]|| + ||E[e_Y(\hat{s}(X) - s^*(X))Xe_Z^2]|| \\
&\leq E[||\hat{e}_Y - e_Y|| \cdot ||X||] + E[||e_Y|| \cdot ||\hat{s}(X) - s^*(X)|| \cdot ||X||] \\
&\leq E[||X||] \cdot ||\hat{\tau}_r - \tau_r^*|| + E[(|Y| + C)\mathbb{1}\{\hat{s}(X) \neq s^*(X)\} \cdot ||X||] \\
&\leq C||\hat{\tau}_r - \tau_r^*|| + C\Pr(|\Delta p^*| \leq 2||\hat{p} - p^*||_\infty) \\
&\leq C||\hat{\tau}_r - \tau_r^*|| + C||\hat{p} - p^*||_\infty^\zeta \\
&= o_p(1) + o_p(1)
\end{aligned}$$

The second inequality holds as e_Z is bounded in magnitude by 1. The fourth uses Assumption 9. The fifth invokes Assumption 5.

Lemma 11 *Suppose that Assumptions 1 - 5 and Assumptions 8 - 9 hold.*

$$E_n[XX^T\hat{e}_Z^2] = E[XX^Te_Z^2] + o_p(1)$$

Proceeding as in the previous Lemmas,

$$E_n[XX^T\hat{e}_Z^2] - E[XX^Te_Z^2] = E_n[XX^T\hat{e}_Z^2] - E[XX^T\hat{e}_Z^2] + E[XX^T\hat{e}_Z^2] - E[XX^Te_Z^2]$$

The first term is $o_p(1)$ by Lemma 7, Assumption 9, and Theorem 2.10.5 of Vaart and Wellner (2023). For the second term,

$$\begin{aligned}
E[(\hat{e}_Z^2 - e_Z^2)XX^T] &= 2E[e_z(\hat{e}_Z - e_Z)XX^T] + E[(\hat{e}_Z - e_Z)^2XX^T] \\
&= 2E[e_zX^T(\gamma^* - \hat{\gamma})XX^T] + E[(X^T(\gamma^* - \hat{\gamma}))^2XX^T] \\
&= E[(X^T(\gamma^* - \hat{\gamma}))^2XX^T].
\end{aligned}$$

The final equality holds as $E[e_Z|X] = 0$. Using the Frobenius norm,

$$\begin{aligned}
||E[(X^T(\gamma^* - \hat{\gamma}))^2XX^T]||_\mathcal{F} &\leq E[||(X^T(\gamma^* - \hat{\gamma}))^2XX^T||_\mathcal{F}] \\
&\leq E[(X^T(\gamma^* - \hat{\gamma}))^2 \cdot ||XX^T||_\mathcal{F}] \\
&= E[(X^T(\gamma^* - \hat{\gamma}))^2 \cdot ||X||^2] \\
&\leq E[||X||^4]||\hat{\gamma} - \gamma^*|| \\
&= C||\hat{\gamma} - \gamma^*|| \\
&= o_p(1).
\end{aligned}$$

The second line holds by the triangle inequality, and because $X^T(\gamma^* - \hat{\gamma})$ is a scalar. For the third line, $||XX^T||_\mathcal{F} = \sqrt{\text{trace}(XX^TXX^T)} = \sqrt{\text{trace}(X^TXX^TX)} = ||X||^2$. The fifth line holds by Assumption 9.

B.5 Proof of Theorem 3

To prove Theorem 3, we first show that Assumption 6 generates a class of $s(X)$ functions which satisfy Assumption 4. Under Assumption 6,

$$\mathcal{S} \coloneqq \{s(X) = 1 - 2\mathbb{1}\{-X^T\delta > 0\} : \delta \in \mathcal{D}\}.$$

The class of functions $\mathcal{F} \coloneqq \{-X^T\delta : \delta \in \mathcal{D}\}$ is a finite dimensional vector space, which is a VC-subgraph class by Lemma 2.6.16 of Vaart and Wellner (2023). By Lemma 2.6.20 (iii), (v) and (vi) of Vaart and Wellner (2023), passing $\mathcal{F}$ through an indicator function with threshold zero, adding a fixed function, and multiplying by a fixed function preserve this property. We thus conclude that $\mathcal{S}$ is a VC-subgraph class.

For the asymptotic distribution, Lemmas 12 and 13 show that the results of Lemmas 3 and 5 hold with Assumptions 6 and 7 replacing Assumption 3. The results of Lemmas 1, 4 and 6 do not directly rely on Assumption 3, nor does the main body of the proof of Theorem 1. Hence, Theorem 3 can be shown by the same argument with the new lemmas replacing the old.

For the consistency of the estimated standard error, only Lemmas 9 and 10 require modification. In particular, the term $E[\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}||e_Y X||]$ appears in both lemmas. This is shown to be $o_p(1)$ as follows. Let ξ_n be a sequence of positive constants that grow at rate $1/4$.

$$\begin{aligned}
E[\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}||e_Y X||] &\leq E[\mathbb{1}\{|x^T\delta^*| \leq |x^T(\hat{\delta} - \delta^*)|\}(|Y| + \tau_r^*)||X||] \\
&\leq E[\mathbb{1}\{|\Delta p^*| \leq C \cdot ||x|| \cdot ||\hat{\delta} - \delta^*||\}(|Y| + \tau_r^*)||X||] \\
&\leq C \Pr(|\Delta p^*| \leq C \cdot ||x|| \cdot ||\hat{\delta} - \delta^*||) \\
&\leq C \Pr(|\Delta p^*| \leq C \cdot \xi_n \cdot ||\hat{\delta} - \delta^*||) + C \Pr(||x|| > \xi_n) \\
&\leq C(\xi_n \cdot ||\hat{\delta} - \delta^*||)^\zeta + C \Pr(||x|| > \xi_n) \\
&= O_p(n^{-\zeta/4}) + o(1) \\
&= o_p(1)
\end{aligned}$$

Lemma 12 *Suppose that Assumptions 1, 2, 5 - 7 and 8 hold.*

(i) $\sqrt{n}E[m^{e_Y}(\hat{s}, \hat{\gamma}) - m^{e_Y}(s^*, \gamma^*)] = -\sqrt{n}E[e_Y s^*(X)X]^T(\hat{\gamma} - \gamma^*) + o_p(1)$

(ii) $\sqrt{n}E[m^D(\hat{s}, \hat{\gamma}) - m^D(s^*, \gamma^*)]) = -\sqrt{n}E[Ds^*(X)X]^T(\hat{\gamma} - \gamma^*) + o_p(1)$

Let ξ_n be a sequence of positive constants that grow at rate $\zeta/2$. For claim (i), proceed as in Lemma 3, obtaining the same decomposition and defining $\mathcal{A}_1$ and $\mathcal{A}_2$ identically. We verify that

$\sqrt{n}\mathcal{A}_1$ and $\sqrt{n}\mathcal{A}_2$ are still $o_p(1)$. For $\mathcal{A}_1$,

$$\begin{aligned}
|\mathcal{A}_1| &\leq E[(|Y| + |\tau_r^*|D)\mathbb{1}\{\hat{s}(X) \neq s^*(X)\}|X^T(\hat{\gamma} - \gamma^*)|] \\
&\leq E[(|Y| + |\tau_r^*|)\mathbb{1}\{|x^T\delta^*| \leq |x^T(\hat{\delta} - \delta^*)|\}||x||] \cdot ||\hat{\gamma} - \gamma^*|| \\
&\leq E[(|Y| + |\tau_r^*|)\mathbb{1}\{|x^T\delta^*| \leq ||x|| \cdot ||\hat{\delta} - \delta^*||\}||x||] \cdot ||\hat{\gamma} - \gamma^*|| \\
&\leq \Big(\xi_n \cdot E[(|Y| + |\tau_r^*|)\mathbb{1}\{|x^T\delta^*| \leq C \cdot \xi_n \cdot ||\hat{\delta} - \delta^*||\}] \cdot ||\hat{\gamma} - \gamma^*|| + \Pr(||x|| \geq \xi_n)||\hat{\gamma} - \gamma^*||\Big) \\
&\leq \Big(C \cdot \xi_n \cdot \Pr(|\Delta p^*| \leq C \cdot \xi_n \cdot ||\hat{\delta} - \delta^*||\}) \cdot ||\hat{\gamma} - \gamma^*|| + \Pr(||x|| \geq \xi_n)||\hat{\gamma} - \gamma^*||\Big) \\
&\leq C \cdot \xi_n^{1+\zeta}||\hat{\delta} - \delta^*||^{\zeta} \cdot ||\hat{\gamma} - \gamma^*|| + \Pr(||x|| \geq \xi_n)||\hat{\gamma} - \gamma^*|| \\
&= O_p(n^{-(1+\zeta^2)/2}) + o_p(n^{-1/2}).
\end{aligned}$$

where the last equality holds as $\Pr(||x|| \geq \xi_n) = o(1)$. As $\zeta > 0$, $\sqrt{n}\mathcal{A}_1 = o_p(1)$.

For $\mathcal{A}_2$,

$$\begin{aligned}
|\mathcal{A}_2| &\leq CE[\mathbb{1}\{\hat{s}(x) \neq s^*(x)\}|\Delta p^*(x)|(|\tau(x)| + |\tau_r^*|)] \\
&\leq CE[\mathbb{1}\{|x^T\delta^*| \leq |x^T(\hat{\delta} - \delta^*)|\}|\Delta p^*(x)|(|\tau(x)| + |\tau_r^*|)] \\
&\leq CE[\mathbb{1}\{|\Delta p^*| \leq ||x|| \cdot ||\hat{\delta} - \delta^*||\}|\Delta p^*(x)|(|\tau(x)| + |\tau_r^*|)] \\
&\leq CE[\mathbb{1}\{|\Delta p^*| \leq ||x|| \cdot ||\hat{\delta} - \delta^*||\} \cdot ||x||(|\tau(x)| + |\tau_r^*|)] \cdot ||\hat{\delta} - \delta^*|| \\
&\leq C\left(E[\mathbb{1}\{|\Delta p^*| \leq \xi_n||\hat{\delta} - \delta^*||\}(|\tau(x)| + |\tau_r^*|)] \cdot \xi_n||\hat{\delta} - \delta^*|| + \Pr(||x|| \geq \xi_n)||\hat{\delta} - \delta^*||\right) \\
&\leq C(\xi_n||\hat{\delta} - \delta^*||)^{(1+\zeta)/2} + C\Pr(||x|| \geq \xi_n)||\hat{\delta} - \delta^*|| \\
&= O_p(n^{-(1+\zeta^2)/2} + o_p(n^{-1/2}).
\end{aligned}$$

The sixth inequality holds by Assumptions 5 and 8. This implies $\sqrt{n}\mathcal{A}_2 = o_p(1)$. The proof of claim (ii) follows its proof in Lemma 3 with the same additional steps as above.

Lemma 13 *Suppose that Assumptions 1, 2, 5 - 7 and 8 hold.*

(i) $\mathbb{G}_n(m^{e_Y}(\hat{s}, \hat{\gamma})) = \mathbb{G}_n(m^{e_Y}(s^*, \gamma^*)) + o_p(1)$

(ii) $\mathbb{G}_n(m^{D}(\hat{s}, \hat{\gamma})) = \mathbb{G}_n(m^{D}(s^*, \gamma^*)) + o_p(1)$

For claim (i), consider $E[(m^{e_Y}(\hat{s}, \hat{\gamma}) - m^{e_Y}(s^*, \gamma^*))^2]$. Repeating the steps of Lemma 5, the only term that must be dealt with differently is $CE\left[(Y^2 + \tau_r^{*2})\mathbb{1}\{\hat{s}(X) \neq s(X)\}\right]$. We show that this

remains $o_p(1)$ under the new assumptions. Let ξ_n be an increasing sequence of positive constants

$$\begin{aligned}
CE\left[(Y^2+\tau_r^{*2})\mathbb{1}\{\hat{s}(X)\neq s(X)\}\right] &\leq E\left[(Y^2+\tau_r^{*2})\mathbb{1}\{|x^T\delta^*|\leq|x^T(\hat{\delta}-\delta^*)|\}\right] \\
&= E\left[(Y^2+\tau_r^{*2})\mathbb{1}\{|x^T\delta^*|\leq|x^T(\hat{\delta}-\delta^*)|,|\Delta p^*|\leq C|x^T\delta^*|\}\right] \\
&\leq E\left[(Y^2+\tau_r^{*2})\mathbb{1}\{|\Delta p^*|\leq C||x||\cdot||\hat{\delta}-\delta^*||\}\right] \\
&\leq C\Pr(|\Delta p^*|\leq C||x||\cdot||\hat{\delta}-\delta^*||) \\
&\leq C\Pr(|\Delta p^*|\leq C\xi_n\cdot||\hat{\delta}-\delta^*||)+\Pr(||x||>\xi_n) \\
&\leq C||\hat{\delta}-\delta^*||^{\zeta}+\Pr(C\xi_n||\hat{\delta}-\delta^*||>\eta)+\Pr(||x||>\xi_n)
\end{aligned}$$

The second line uses Assumption 7 (ii), and the fourth line uses Assumption 8. In the final line, η is as defined in Assumption 5. Under Assumption 7, $||\hat{\delta}-\delta^*||=O_p(n^{-1/2})$. This implies that, with $\zeta>0$, the first term is $o_p(1)$. Furthermore, we can construct ξ_n such that $\xi_n\to\infty$ and $\xi_n||\hat{\delta}-\delta^*||=o_p(1)$. Such a choice renders the last two terms $o(1)$. It follows that $E\left[\mathbb{1}\{\hat{s}(X)\neq s(X)\}\left(D(Z-X^T\gamma^*)\right)^2\right]=o_p(1)$. The proof can then proceed as in Lemma 5. The argument for claim (ii) is identical.

C Monte Carlo Calibration Details

Parameter values are set as follows. As described in Section 4, δ_0 and $||\delta_1||$ are set to match the population shares of compliers and defiers. Given δ_0 and $||\delta_1||$, $\frac{\beta_1^T\delta_1}{||\delta_1||^2}\tau_x^*$ and $(E[U(1)V]-E[U(0)V])\tau_u^*$ are chosen to fix τ_r^*. Note that τ_x^* and τ_u^* are determined by δ_0 and $||\delta_1||$. Hence, we use τ_r^* to fix $\frac{\beta_1^T\delta_1}{||\delta_1||^2}$ and $(E[U(1)V]-E[U(0)V])$.

The remaining parameter values are determined as follows. As X is normally distributed with mean zero, the vector $(X_1^T\delta_1, X_1^T\beta_1)$ is so too. Rather than fix δ_1 and β_1, we hold this distribution fixed. To fix the covariance matrix, we require an additional condition, so we set $\frac{\beta_1^T\delta_1}{||\beta_1||\cdot||\delta_1||}$ at -0.5. In each simulation, we draw δ_1 and β_1 such that the population covariance matrix satisfies these conditions.

Finally, we require the covariance matrix for $(U(1),U(0),V)$. The value of $E[U(1)V]-E[U(0)V]$ provides one restriction, and the variance of V is normalised to one without loss of generality. Beyond this, we impose that the correlation between $U(1)$ and V is -0.5, the correlation between $U(0)$ and V is 0.5, the correlation between $U(0)$ and $U(1)$ is -0.25, and that the variances of $U(0)$ and $U(1)$ are equal. Under these restrictions, the variances of $U(0)$ and $U(1)$ are adjusted to attain the desired τ_r^*.

In the design with no defiers, we require an additional condition to fix the variance of $X_1^T\beta_1$. We fix this to 1.